

Háskoli Islands

Liffræðiskor

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Liffræði

LIFE HISTORY OF SCAD MACKEREL
***Decapterus macarellus* CUVIER, 1833, IN THE**
WATERS OFF THE CAPE VERDE ISLANDS

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Summary

The capeverdean scad mackerel - *Decapterus macarellus* Cuvier, 1833, is a small pelagic, schooling fish, native to the Cape Verde Islands, which generally does not grow to more than 35 cm FL and 250 - 300 g in weight. In most cases, it matures and spawns at 3 - 4 years of age at a length of 19 - 22 cm FL. The present work deals with the capeverdean mackerel stock, its biology, and exploitation. The aims of the present work are to estimate biological parameters of the *Decapterus macarellus* from the Cape Verde Islands and to assess the state of the stock and the fishery. The study is based on research carried out since 1983. Growth parameters, the proportions at age-groups, and the mean length at age-groups were estimated by using an efficient method of analyzing distributions mixtures (MacDonald and Pitcher method) : $L_{inf} = 30.1$ cm FL ; $K = 0.34$ year⁻¹ and $t_0 = 0.11$ per year. The weight-length relationships were estimated for the immature portions of the stock, for female, and for male, as well as for all sexes . The weight-length relationship for all sexes was used in stock assessments. The maturity ogive equation was estimated. The instantaneous rate of natural mortality was $M = 0.64$ year⁻¹, estimated by Pauly's method, and $M = 0.43$ year⁻¹ estimated by Tanaka's method. A value of 0.4 was adopted and used in the assessment. Study of the diet and feeding habits of the capeverdean scad mackerel indicates that it feeds almost exclusively on copepods and decapod larvae. It feeds mostly during daylight. The larger fish can have a more diverse diet which includes larger prey (euphausiaceans). In order to estimate the relative abundance and fishing mortality coefficients, two methods of VPA were applied: the Length-cohort analysis and the age-cohort analysis. Due to the difficulties with the CPUE data, the tuning of the cohort analysis could not be applied. Based on a selection pattern, estimated from the age cohort analysis, the yield per recruit was estimated. The biological reference points were calculated. In order to estimate the yield potential, the stock recruitment relationship was investigated based on Shepherd the method (1982). The maximum sustainable yield was estimated to be equal to 5 - 6 thousand tons. Since the current catches are around 3 thousand tons, it was concluded that there is room for increasing the catch from a level of 3 thousand tons to 5 - 6 thousand tons without putting the stock at risk. A stochastic simulation model based on 100 Monte Carlo simulations with fluctuations in the recruitment and in the fishing mortality rates, were carried out in order to investigate what kind of development is likely and unlikely to happen to the catch and to the spawning stock biomass of the capeverdean scad mackerel until the year 2005. Finally, it was concluded that it is necessary to improve the data collection.

Key words: *Decapterus macarellus*, Cape Verde Islands, biology, stock assessment

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1. Introduction

1.1 General overview of the species

Decapterus macarellus Cuvier, 1833 (family Carangidae) is a small pelagic, schooling fish, native to the Cape Verde Islands, the Island of Santa Helena, the Gulf of Mexico, the Gulf of Guinea, and the northern part of the Gulf of Aden, which lives in water columns from the surface down to 200 meters (Lagler *et al.*, 1977 and Fisher *et al.*, 1980).

The scad a mackerel is a relatively small fish with mature individuals ranging between 17 and 33 cm from ages of 2 - 7 years¹. The scad mackerel has an elongated body. The length is five to six times that of the height (Fisher *et al.*, 1980 and Jardim, 1995). Above the lateral line, the scad mackerel appears dark grayish in color while below it the sides are light grayish colored. The scale margins are not dotted with black specks, and have only a black speck over the operculum (Fig. 2).

Fisher *et al.* (1981) describe this species as having a row of black dots along the lateral line, 29 to 30 rays in the dorsal fin and 25 to 30 rays in the anal fin. The scales are small and hard. The number of scales on the lateral line ranges between 77 to 98. The teeth in the jaws and on the roof of the mouth, as well as the tongue, are small. The inferior jaw is somewhat angular and the gill filaments range in number from 34 to 38. There are no signs of sexual dimorphism in this species.

The scad mackerel, unlike other carangids, is poorly studied. Little is known yet about the reproduction pattern of scad mackerel (*Decapterus macarellus*). The capeverdean scad mackerel probably spawns in water columns near the beaches around the islands from April to July².

Little is known of the biology of the capeverdean scad mackerel larvae. Povocinikov *et al.* (1991) conducted an experiment by keeping the scad mackerel eggs caught during the ichthyoplankton surveys during May 1981 and September 1985 in the northern Gulf of Aden in an aquarium at temperatures ranging from 27.8 to 29.5 degrees during some days. They found that the hatching occurred after 24 hours with larval length at hatching ranging from 1.5 mm to 2 mm. The water temperature during the spawning period of *Decapterus macarellus* in the water off the Cape Verde island ranges from 23 to 25 degrees, so one could imagine that the hatching

¹ The age-groups are estimate from the length-frequency data.

the larva of capeverdean *Decapterus macarellus* could occur later than 24 hours after spawning, based on the principle that the hatching is proportional to water temperature (with some limits) (Longhurst and Pauly, 1987).

The scad mackerel seems to feed mostly on planktonic invertebrates³, either holoplankton or meroplankton. They seem to be active feeders during the daylight near the surface and start migrating to deep water after sunset for resting⁴.

Decapterus macarellus is recognized by different local names in different places (Jardim, 1995). The most commonly known are in Cape Verde as "cavala preta", in Portugal as "charro-olho-largo", in England as "scad mackerel", in France as "Maquereau", and in Spain as "macarella caballa".

Depending on location and society, the scad mackerel have been eaten by man and used as bait in various fisheries.

This species is well known to the inhabitants of the Cape Verde Islands as "cavala preta". Together with the carangid *Selar crumenophthalmus*, they make up the most frequently landed species in the Cape Verde Islands either by artisanal fleet or by the industrial fleet (Anon, 1994).

Today capeverdean scad mackerel has acquired at least a good deal of the attention it deserves. It is not only recognized as a valuable bait for the tuna fishery, but also as an important source of protein for most inhabitants of the Cape Verde Islands with scarce financial resources, and also as a source of income. It is recognized, also, as an extremely important part of the diet of commercial species in the water off the Cape Verde Islands, but whether the changes in its abundance could affect the abundance of the commercial species, is still not well studied.

Usually the capeverdean scad mackerel is caught mainly by purse seines and in small quantities by beach seines.

1.2 Objectives

The aim of this paper is to study the life history, age structure, and sex ratio of *Decapterus macarellus* in the waters around the Cape Verde Islands, and more importantly evaluate the stock-dynamics and assessment of the scad mackerel in the area.

² April-July seems to be the main spawning season, although capeverdean scad mackerel and others tropical fish alike spawn throughout the year.

³ Little is known about the feeding habits of the *Decapterus* sp.

⁴ Based on paper by Hales (1987) and on personal observations.

2. The environmental conditions of waters off the Cape Verde Islands

The archipelago of Cape Verde is located 600 km off the coast of Senegal and consists of ten Islands of which nine are inhabited. The total land area of Cape Verde is 4 033 Km². The continental shelf is irregular and narrow and from 0 to 200 m depths it has an area about of 5 394 Km². The shelf plus slope extend down to 500 m, and constitute an area of about 7 650 km² (Anonymous, 1993). See Figure 1 in appendix.

There are two climatic seasons in the Cape Verde Islands: warm season (wet season) from July to November and cold season (dry season) from December to June depending on the NE winds and on air temperature (Almada, 1993). During the cold season the surface temperature of the ocean is around 21-23°C, while during the warm season the surface temperature is around 25-27°C.

The main current system is the Canary current which runs southwesterly along the northwest African Coast, but beyond this area it turns west and northwest and then becomes the Northern Equatorial current. The average speed of the residual current, through the Cape Verde archipelago, is less than 0.5 knots (Stomme and Sundby, 1981 and Almada, 1992) but the current speed is variable in time and between areas and its strength can, in places, make fishing difficult. Near the banks of Nova Holanda, Joao Valente and Noroeste de S. Antao, there seems to be some up-welling of deep water at the western edge and the opposite process, the down-welling, seems to occur near the eastern edge (Almada, 1992). The up-welling is strongest during the cold season.

Little is known of the temporal pattern of productivity of phytoplankton and zooplankton in the area.

3. The fishery

Probably, the capeverdean scad mackerel fishery started in the early 1930s as a complement of the tuna fishery (Carvalho, M. E and Caramelo, A. M., 1996 and Almeida, 1996). Originally, the capeverdean scad mackerel was caught as bait for the tuna fishery and as a source of food for the inhabitants of the Cape Verde Islands.

Statistical reports on the catches of the capeverdean scad mackerel, as well as for other species, were initiated in the early 1980s. Reports on the catches show that

from the early 1980s till 1992 the catches fluctuated from 200 tons to 1000 tons with exceptions in 1985 and 1989 (see Fig. 6). In 1992, the scad mackerel fishery abruptly increased to almost twice the yield of 1990 and has been increasing until the present (Fig. 6). This was mainly associated with the partial interruption of tuna fleet activities and by the appearance of two Japanese freezing vessels with large capacities for freezing which have been buying capeverdean scad mackerel caught by the local fleet. One could say that the sudden development of the scad mackerel fishery is mainly due to the market demand.

The predominant fishing gears used in the fishery for the capeverdean scad mackerel are beach seines (Anon. 1985) which are operated from boats ranging from 4 to 6 meters in length with or without outboard motors (artisanal fleet) and purse seines (Anon. 1985) operated from the boats with inner motors ranging from 7 meters up to 22 meters (Industrial fleet). The major portion of the catches are made by purse seines (90% of the catch). Catches are taken throughout the year, although more intensively in the warm season (from July to December) during the recruitment to the fishery. The fishery is confined to within approximately 5 miles of the coast.

Historically, the major fishing area was around the Islands of S. Antao, S. Vicente, and S. Nicolau with the main landing points in the islands of S. Vicente and S. Nicolau. Actually, the capeverdean scad mackerel fishery is based around most of the island with landing places mainly in the islands of S. Vicente and S. Tiago.

4. Material and methods

Length, weight, sex ratio, and maturity stages measurements

Regular sampling, in order to record the length of capeverdean scad mackerel, (*Decapterus macarellus*) started in 1983 in the fish market of S. Tiago Island and at the main fishing ports of S. Vicente Island. In principle, at least 30 specimens were measured every month at the above named places.

Over the next few years, the sampling was gradually intensified and in 1988 sampling on length, weight, sex, maturity stage and weight of gonads of 60 specimens per month started in operation at the laboratory in the Fisheries Research Institute at S. Tiago Island (ex-INIP) and at the laboratory of a filial of Ex-INIP at S. Vicente Island.

In 1989, the sampling on length measurements was extended to other islands by placing Enumerators on the main fishing ports or at the main landing places. In the mean time, the enumerator recorded and still records the catch per species per boat, the effort in fishing days (for industrial fleet) and in sea trips (for the artisanal fleet). These data are further analysed at The Fisheries Institute (Now called INDP) in order to establish total catch and effort per species, per fishing gear per month, per island, and per year (Anon, 1993).

All measurements were carried out on fresh specimens of scad mackerel that were frozen shortly after capture. All the samples were taken from the commercial catches.

Recorded fish lengths are in centimeters. These represent the fork length measured from the tip of the mandible to the deepest part of the forks. All weights are in grams.

The sex and maturity stages were determined visually without the use of a microscope. The maturity stages run from I to VII (See section 5.7).

The gonad weight of the female scad mackerel and the weight of male testes have been recorded with the precision of 0.1 gram.

With the above qualifications, a data base comprising 28087 fish-lengths is available covering the period from 1983 to 1996 and including 4762 individual measurements of weight, sex, maturity stage and gonad weights are available from 1988 to 1996.

Stomach contents

In 1996, specimens were collected for study of stomach contents off the island of S. Vicente. In all, 242 stomachs were collected: 100 during the cold season (January - February) and 142 during the warm season (June - August). 30 specimens were collected during early morning, 61 during midday and 51 during the night. The stomachs were preserved in 5 % buffered formalin. After preservation for some days, stomachs were washed and then preserved in 70% isopropanol. The prey items were separated, the content was weighed (to nearest 0.01 g), individuals counted and identified to order, genus or species when possible, and then preserved in 70% isopropanol. All the specimens were measured (Fork Length (FL) , in cm), weighed (nearest gram) and sexed before the gastrointestinal tract was removed.

Statistical analyses.- The data sets for the cold season (January - May) and for the warm season (June - December) were separated by length size and by time period categories and analyzed separately. Percent frequency of occurrence (%FO) was

computed by counting the number of stomachs in which an item occurred, dividing by the total number of stomachs examined and multiplying by 100. Percent number (%N) was calculated by dividing the total number of each item by the total number of all items and multiplying by 100. Percent weight (%W) was computed by dividing the total weight of each item by the total weight of all items and multiplying by 100. The total relative weight of food in stomachs (TRWF) was calculated as the weight of food in stomachs divided by the weight of fish. The long term food diversity (LFD)⁵ or food diversity potential was defined as $S_c(\ln n_c)^{-1}$, where S_c is the cumulative number of food species, and n_c is the cumulative number of food animals (Mattson, 1990).

Age measurements

Otoliths were dissected from 200 specimens of *Decapterus macarellus* of different sizes from the S. Vicente area, in order to evaluate the age of the individuals (see chapter 5.1). Standard methods were applied for determination of annual rings by the author under the guidance of qualified personnel from the MRI, but no significant information was obtained.

Catch in number by length and by age-groups

In order to compute the number of landed capeverdean scad mackerel in each length group for each year from 1983 - 1996, the first thing to be done is to change the amount of landed scad mackerel in tons into some sort of count, for example, the total number of landed scad mackerel. For that, the mean weight for each year was calculated based on an initial weight-length relationship. Such relationship is treated in chapter 5.4.

From the weight-length relationships shown in chapter 5.4, it was possible to compute the mean weight across all fish in each year :

$$\overline{W}_y = \frac{\sum_i \alpha \cdot l_i^\beta L_i}{\sum_i L_i}$$

where, l is the frequency, L is the length category, and α and β are coefficients which can be computed using regression analysis.

⁵ The long-term food diversity is well described in Mattson (1990)

When the mean weight had been calculated, it was possible to compute the total number of landed scad mackerel. This was obtained by dividing the average weight into the total catches in tons (Y_y) for each year :

$$C_y = \frac{Y_y}{\bar{W}_y}$$

Then the catch in number could be determined by using the proportions at length, which are calculated in chapter 5.3.1 based on the distribution mixture method :

$$C_{ly} = C_y P_{ly}$$

The proportions at length and the catch in number at length are available in Table 2 and Table 5 in the appendix. The catch in number at length have been computed in order to be used as a Length's Cohort Analysis (Jones's Cohort) input.

The catch in number at age was computed by following the same procedures as for catch in number at length. Here, this is done by using the earlier obtained number on the proportional, split at age which had been calculated in chapter 5.3.1 based on the distribution mixture method :

$$C_{ay} = C_y P_{ay}$$

Table 2 in the appendix shows the proportions at age, the weight per year, the total catch in number of landed scad mackerel in each year (1983 - 1995) and the catches in number at age in each. The catch in number at age had been computed in order to be an input to the age Cohort Analysis.

It should be pointed out here that difficulties encountered during the computation of proportions at age (see section 5.3) due to inadequate sampling, have been carried over into catches in number at age computations. Hence, in some age groups, there are some anomalous catches in number that could be avoided if the sampling had been adequate.

Total catch in tons (Landings) and effort data

Total catches in weight and effort data are available by artisanal fleet and industrial fleet and years for 1983 - 1995. The effort data for the artisanal fleet and industrial fleet are expressed in the number of sea trips and in fishing days, respectively (Table 1).

Both the artisanal fleet and the industrial fleet have used the same fishing gear (purse seines), hence the sum of the total catches of both fleets have been used in further assessments. The effort data were standardized in fishing days. The total catches in tons and total effort in fishing days are described in Table 1 and in Figure 6 in the appendix. As we see in Figure 6 in the appendix, the total catches and total effort vary from 1983 to 1995 showing no trends.

Environmental data

Environmental data (wind speed, wind direction, air temperature) from 1985 - 1995 has been obtained from the Meteorological Center in the Cape Verde Islands. Data of sea surface temperature from satellite surveys has been obtained from CRODT (Centre de Recherches Oceanographiques du Dakar-Tiaroy). This data series starts in 1985.

In addition, the most appropriate methods in fish stock assessment have been used to evaluate the state of this stock (Stefánsson, 1994 and Jakobsson *et. al.*, 1993).

5. Biology of *Decapterus macarellus*

5.1. Growth parameters

The study of growth means basically the determination of the body size as a function of age. Therefore, most all stock assessment methods work with age composition data. In temperate waters, such data can be obtained by the counting of annual (winter) rings in the hard parts such as scales or otoliths. But in subtropical and tropical waters, it is very difficult to distinguish year rings, because drastic changes in the environment do not occur. Recently, work has started on reading daily rings to count the age of the fish in number of days, but this method has only been applied to very young fish.

In spite of the complication and expense of obtaining the age data, a large number of methods has been used by biologists to estimate the average size and relative abundance of component age groups in samples. The most commonly used depend on some sort of graphical interpretation. The general principle is to estimate the parameter of a particular component age-group from a region of the overall length distribution, where the effects of the others components are negligible.

It is common in practice, and often reasonable, to assume that the component size as to be normal distribution, or easily transformed normal, and then move from one side of the size-frequency distribution to the other, graphically identifying and removing the components in turn. Various methods of this type are described by Tanaka (1962), Taylor (1965), Bahattacharya (1967).

As Rosenberg *et. al.* (1988) pointed out, these above referred to methods are sometimes useful in the absence of a computer. They are almost subjective and have been superseded by the numerical methods of Slicing, Macdonald and Pitcher, and others (Macdonald and Pitcher, 1979).

In our case, the growth parameters are estimated from the data on length frequencies from the samples of commercial catches indicated in chapter 5.2. In this study, the age-groups are estimated by making use of a versatile and efficient method of analyzing distribution mixtures (Macdonald and Pitcher method).

5.1.1. Macdonald method

The MacDonald Method is one of the numerical methods developed for the resolution of length-frequencies into age compositions, culminating with maximum-likelihood methods. This method estimates the mean length at age (see Macdonald and Pitcher, 1979).

For this method, it is necessary to have an assumption concerning the distribution of length-frequencies and less critically, some guess of the number of cohorts. Then, constraints are required in order that the mean growth follows some chosen apriori growth function (that could be the Von Bertalanffy function) and the variances follow some simple pattern.

The SSE (Sum of Squared Errors) is the sum of squared differences between the observed and calculated frequencies over the entire range of the sample. The SSE determines the set of parameters (L_{inf} , K , t_0 , and the proportions in each age-group) which minimize the SSE.

$$SSE = \sum_1^n SSE_y$$

$$SSE_y = \sum_{L_m}^{L_M} (\ln(L_{obs}) - \ln(L_{est}))^2$$

where the index "y" is year.

For capeverdean scad mackerel (*Decapterus macarellus*), we assumed that the length frequencies of each year from 1983 to 1995 are log-normal distributed, that the number of cohorts (components in the mixture) is seven, based on the Bhattacharya analysis made by Carvalho (1993) and by Jardim (1995) on the same data of the same species, and that the mean growth follows the Von Bertalanffy function ($L_t = L_{inf}[1 - \exp(-K(t - t_0))]$). Based on the above assumptions, a worksheet was prepared in the Microsoft spreadsheet Excel. Initial values of growth parameters (L_{inf} , K , and t_0) and proportions at age-groups are guessed at. An interactive process was been started in order to find the best solutions. The interactive process stops only when the SSE reaches a global minimum.

In the appendix, in Table 2 - 3 are shown the growth parameters along with the proportions at age-groups, the mean length at age, the variances within ages groups, and the mean weight at age best fitted by the above method. The SSE for each year was calculated and the sum of the SSE reached its minimum at 52. The observed and the estimated frequencies are shown graphically in Figure 3. For all years, except for 1988 and 1991, the observed and estimated frequencies are well fitted. Although the observed and estimated frequencies are not fitted as well as in other years, the residuals are low values and both observed and estimated curves seem to have the same shape.

For the capeverdean scad mackerel the best estimates of growth parameters are the following:

$$L_{inf} = 30.1 \text{ cm} ; K = 0.34 \text{ per year} ; t_0 = 0.11 \text{ year}$$

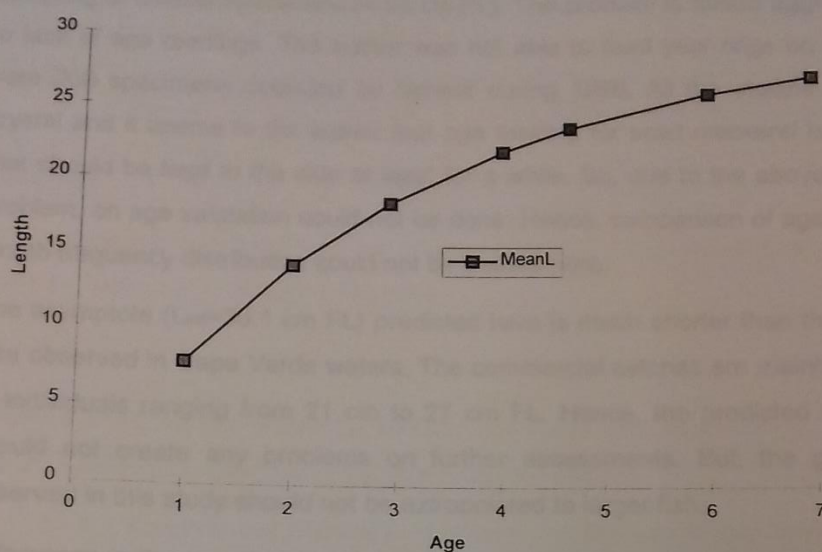


Figure 5.1.1 . - Von Bertalanffy growth curve for *Decapterus macarellus*. Length in cm FL and age in years.

Comparing the growth parameters estimated here to those estimated by other authors, the present values appear close to those estimated by Jardim (1995).

Table 5.1.1 - Estimation of growth parameters for the capeverdean scad mackerel (*Decapterus macarellus*).

Reference	Linf (cm)	K (per year)	to (year)	Method
Carvalho (1993)	32-36	0.44-0.54	-	ELEFAN81
Jardim (1995)	31.5	0.43	-1.56	Stammatopoulos
Caramelo et. all (1995)	41	0,43	-	FISAT1.11 95
This study	30.1	0.34	0.1	MacDonald

5.1.1.1 Discussion

The study of growth parameters of caperverdean scad mackerel resulted in a lot of difficulties, while establishing valid estimates.

Difficulties resulted due to inadequate data collection. One could see that from 1983 to 1993 (see Table of proportions at age), there is clearly an oversampling of specimens ranging from 23 to 27 cm in fork length, while there is insufficient

sampling of smaller specimens (< 23 cm FL). The problem is further aggravated due to lack of age readings. The author was not able to read year rings on the otoliths from 200 specimens collected by himself during 1996. All the otoliths looked like crystal and it seems to the author that age reading for scad mackerel is something that should be kept to the side at least for a while. So, due to the above mentioned problem, an age validation could not be done. Hence, comparison of ageing with the length frequency distribution could not be treated here.

The asymptote ($L_{inf}=30.1$ cm FL) predicted here is much shorter than the maximum size observed in Cape Verde waters. The commercial catches are mainly composed of individuals ranging from 21 cm to 27 cm FL. Hence, the predicted value of L_{inf} should not create any problems on further assessments. But, the growth rates observed in this study should not be extrapolated to larger fish.

Differences in the growth rates observed in the Jardim's study, the Caramelo's study and the study herein, are believed to be due mainly to methodology, since all studies are based on the same data.

Finally, although the method here used relied on some assumptions, it seems to be robust. The distribution mixture methods like the one used here have the advantage of being justifiable on statistical grounds. We have some idea of their behavior when the assumptions of the models are met, and they can potentially yield estimates on the variance of estimators and the sensitivity to the particular characteristics of the data.

5.2. Weight - length relationship

Data from the weight - length relationship of capeverdean scad mackerel can be calculated, and are available on a monthly basis over the whole range of years from 1988 to 1996. A summary of the data set is given in Table 5.2.1.

In order to calculate seasonal changes in the weight - length relationships among capeverdean scad mackerel, seasonal regressions of weight on length representing the warm season (July - December) and the cold season (January - June) were separately computed from commercial samples for both sexes as well as immatures (<19 cm FL), spanning the 1988 - 1995 period. In addition, the weight - length relationship was calculated for both sexes for the whole year from the pooled data.

Table 5.2.1. - The number of *Decapterus macarellus* on which calculation of weight at length by sex and season in 1988 - 1995 were based.

Season	Immature		Female		Male		All sexes	
	Number	L. range	Number	L. range	Number	L. range	number	L. range
Jan. - Jun.	99	17-19	1074	20-32	498	20-32	1634	19-32
Jul. - Dec.	250	13-19	892	20-31	400	20-32	1542	19-32
Whole year	349	13-19	1966	20-32	889	20-32	2797	19-32

The regression coefficients for all sexes, each sex, immatures and seasons are given in Table 5.2.2, using the equation

$$W = \alpha L^{\beta}$$

where w and L are the weight (g) and the Fork length (cm) respectively, and α and β are the coefficients determined by regression techniques.

Table 5.2.2. - Length/weight regression coefficients for the capeverdean scad mackerel by season and whole year based on equation described in section 5.4.

Seasons	immatures		Females		Males		All sexes	
	$\alpha \cdot 10^{-2}$	β	$\alpha \cdot 10^{-2}$	β	$\alpha \cdot 10^{-2}$	β	$\alpha \cdot 10^{-2}$	β
Jan - June	1.01	3.05	0.68	3.19	1.10	3.03	0.78	3.14
July. - Dec.	0.50	3.28	1.09	3.04	2.86	2.74	0.68	3.18
Whole year	0.46	3.31	0.88	3.11	1.47	2.94	0.70	3.17

Table 5.2.3. - The calculated weight (g) of length for capeverdean scad mackerel by season and sex where I, F and M denote immatures, females and males at the most frequent lengths of 19 and 25 cm respectively.

Season	I (19)	F(25)	M(25)	All sexes(25)
Jan. - Jun.	79.8	194.6	190.6	193.2
Jul. - Dec.	77.3	191.6	192.7	191.3
whole year	78.7	193.3	191.5	192.3

The immatures seem to be in their prime condition throughout the year and the seasonal changes are negligible (see Table 5.2.2). The mature portion of the stock seems to be in their prime physical state throughout the year and the changes are on the order of only about 2 grams. However, care should be taken when interpreting these results, because the number of observations of sex and by season used in

regression analysis are different, as we see in Table 5.4.1.. Although, all length groups are represented in both seasons.

5.2.1. Discussion

The relation between fish length and weight is much used in fisheries biology for converting catch weights to numbers of fish for further use in stock diagnoses and projections. As growth may be defined as changes in length and weight over time, these parameters are interrelated and often used in attempts to describe the physical state of the fish (Vilhjálmsen, 1994).

There are no changes detected in the physical state of the capeverdean scad mackerel throughout the year, suggesting that the capeverdean scad mackerel feeds heavily throughout the year (see chapter 5.6). However, it must be said here that nothing is known of the fat contents of capeverdean scad mackerel, nor of the feeding habits of this species (at least until now). As we know (see chapter 5.6), the capeverdean scad mackerel feeds heavily on holoplankton and meroplankton, but until now there have been no studies done on seasonal abundance of holoplankton and meroplankton in the water off the Cape Verde Islands which could be useful in interpretation of the changes in the condition of either the immature or mature portions of the stock of capeverdean scad mackerel.

Table 5.2.1.1. - Weight-length regression coefficients for capeverdean scad mackerel

Reference	$\alpha \cdot 10^2$	β	No. of specimens
Carvalho (1987)	1.81	2.87	?
Magnusson&Magnusson (1987)	0.78	3.14	120
Jardim (1995)	0.65	3.17	1441
This study	0.70	3.17	4762

In Table 5.2.1.1 are shown the weight-length regression coefficients found in different studies. It seems to the author that the differences are due to the number of specimens used in the weight-length regression analysis.

5.3. Sex ratio

Sampling for sex ratio was carried out for each length interval indicated in Table 5.3.1. Individuals were obtained from samples of the commercial catches (see chapter 5.2.). In all, 2855 specimens larger than 15 cm were analyzed for sex ratio in the period from 1988 to 1995. Differences were found in seasonal and annual sex ratios, as we see in Tables 5.3.1. and 5.3.2.

Table 5.3.1. - Seasonal sex ratio of capeverdean scad mackerel in the period from 1988 to 1995.

season	Jan.-June				Jul.-Dec.			
L- range	<20cm	[20-25]cm	]25-30]cm	>30cm	<20cm	[20-25]cm	]25-30]cm	>30cm
Males (%)	52	20	30	(33)	16	34	30	(0)
Females (%)	48	80	70	(67)	84	66	70	(100)
Mal+Fem. (n)	89	810	600	(12)	96	590	604	(2)

Table 5.3.2. - Annual sex ratio of capeverdean scad mackerel in the period from 1988 to 1995.

Sex\Range	<20cm	[20-25]cm	]25-30]cm	>30cm
Males (%)	33	32	30	(43)
Females (%)	67	68	70	(57)
Mal.+Fem.(n)	185	1441	1207	(14)

There is a clear prevalence of females in the main size categories landed by commercial fishing boats (20 - 25 cm and 25 - 30 cm). In these size categories the ratio is close to 1 male per 3 females.

It is generally presumed that for fish, the initial sex ratio is genetically determined as 1 : 1. If that is the case, the sexes should occur in equal number in the immature portion of the stock. Unfortunately, like other subjects, this has not been ascertained for capeverdean scad mackerel. The sampling bias is one of the things that could explain the big differences in sex ratio observed in the above referred to length categories. However, future research should be carried out in order to explain these differences.

5.4. Maturity stages

Gonads of *Decapterus macarellus* have an external shape and general structure similar to those of *Sardinella aurita* (Fontana *et. al.*, 1969). To assess the condition of the sexual maturation in the specimens of *Decapterus macarellus*, initially a maturity key of seven stages (Fontana *et. al.*, 1969) was used on a routine basis in a monthly analysis. The stage of maturity of *Decapterus macarellus* was determined visually without the aid of magnifying devices. It should be pointed out that the use of any macroscopical key can be undergo a certain amount of subjectivism, and, also, that the discrimination between two contiguous maturity stages presents difficulties at times. The maturation scale is presented in the appendix in the table 6.

To obtain a more consistent interpretation of the basic data, all observations obtained in the period from 1988 to 1996 were considered in the following manner :

Monthly observations in the period from 1988 to 1995 (2808 fish - from 15 - 32 cm length) were considered for both sexes combined, in order to determine monthly frequencies of maturity stages by length and the maturing ratio which would be used in further assessment in calculation of spawning stock biomass;

Monthly observations in the period from January 1996 to October 1996 separated by sexes and both sexes combined (1404 for the sexes combined, 480 males and 665 females) were used for maturing ratio calculations in order to compare to those calculated from the data in the period 1988 - 1995.

The maturity at length was calculated by using the following formula :

$$P = 1 / [1 + \exp(S_1 + S_2 L)] \quad (\text{Sparre and Venema, 1992})$$

P is the proportion of mature stock, L is the length and S_1 and S_2 are coefficients which can be calculated by regression analysis (the procedures are available in Sparre and Venema, 1992). For both sexes combined S_1 and S_2 were estimated as 9.39 and -0.47 respectively.

5.4.1. Length at maturity

The length at which 50% of individuals are mature is a commonly used reference in fish when dividing stocks into their mature and immature portions.

Table 5.4.1.1. - The mean length (cm FL) at maturity of female, male and both sexes combined of capeverdean scad mackerel in 1988 - 1995 and 1996.

Years	Males	Females	both sexes
1988 - 1995	-	-	19.89
1996	21.78	21.44	21.80

In Table 5.4.1.1 are shown the mean length at maturity of female and male capeverdean scad mackerel in the period from January to October 1996, and both sexes combined in the January - October 1996 and in the period 1988 - 1995.

By analyzing the results in the above referred to table, we conclude that when the capeverdean scad mackerel starts maturing it has a fork length 19 cm.

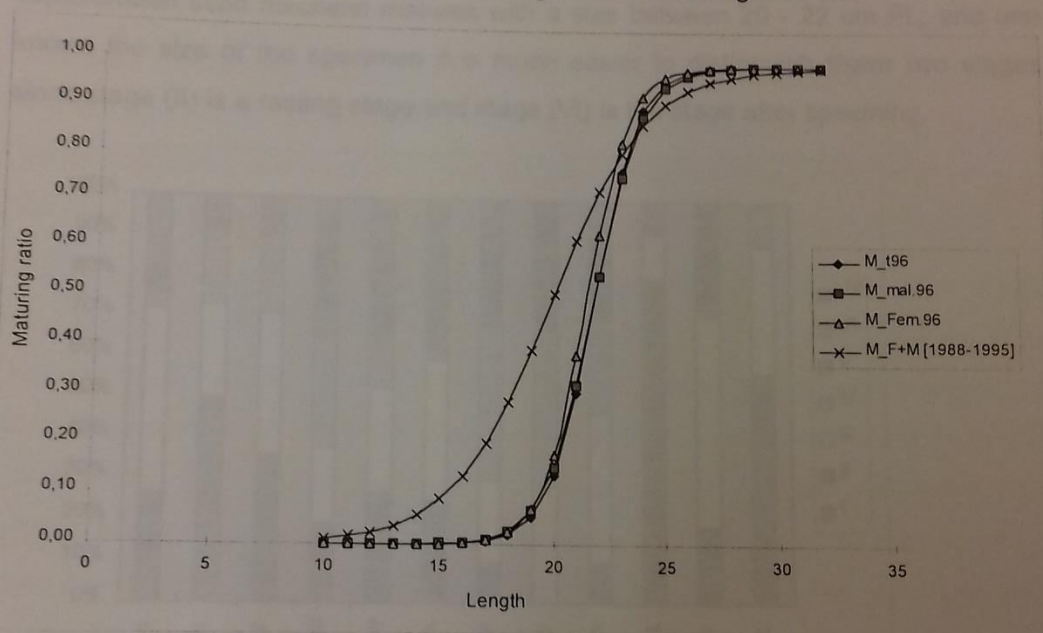


Fig. 5.4.1.1 - Average maturity ogives fitted in the period 1988 - 1995 and in the period Jan. - October 1996 for females, males, and both sexes combined for capeverdean scad mackerel.

As we see in Figure 5.4.1.1, the maturity ogives in the period January - October 1996, for the females, males, and both sexes combined have little differences. The author decided to use the average maturing ratio calculated from the observations in the period from 1988 - 1995. It seems to author that maturing ratio at length is more representative, since it is based on more observations.

5.4.2. Frequency distribution of maturity stages by length and per month

In Figure 5 in the appendix, the stage (I) shows a peak in the size categories of 15 cm FL. In stage (II), the distribution is more dispersed through size categories ranging from 18 to 29 cm FL.

In stage (III), the distribution starts from size 18 and has a peak on 25 cm FL. Stages (IV) and (V) show a normal distribution with both having a peak at size 26. Stage (VI) also shows the peak at size 26 and stage (VII) at the size 24.

Care should be taken when analyzing stage (II) because sometimes it is difficult to distinguish stage (II) from stage (VI). The person in charge of data collection should be an experienced one in order to be able to distinguish between them. If the capeverdean scad mackerel matures with a size between 20 - 22 cm FL, and one knows the size of the specimen it is much easier to distinguish these two stages since stage (II) is a resting stage and stage (VI) is the stage after spawning.

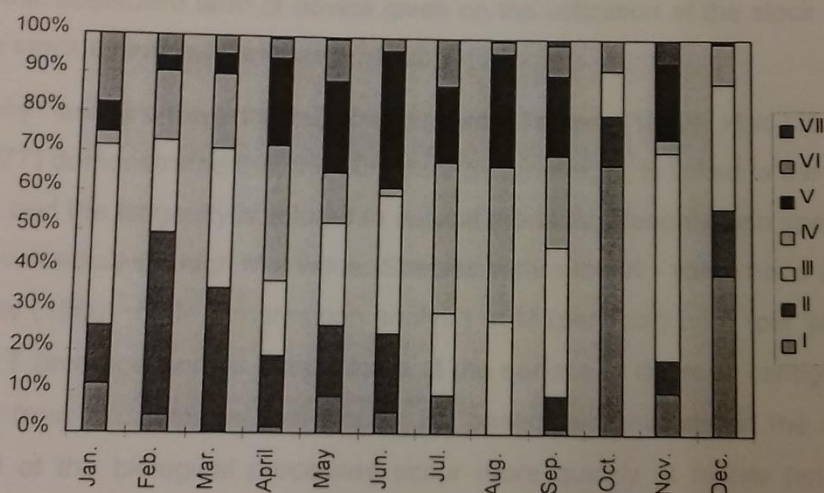


Fig. 5.4.2.1 - Monthly frequency distributions of maturity of capeverdean scad mackerel in the period from 1988 - 1995

Finally, by analyzing the distribution in stage (I), one could conclude that the size 21 cm FL is the line dividing the immature and mature portions of the stock.

Usually, the frequency distributions of maturity per month are very useful in determination of the spawning season. Figure 5.2.2.1 shows the frequency distributions of maturity per month of capeverdean scad mackerel. As is shown in the figure stage (V) occurs mainly from April to September with a peak at June. Since

stage (V) corresponds to the spawning stage, one could say that the capeverdean scad mackerel spawns from April to September. Like other tropical species the capeverdean scad mackerel has a long spawning season.

5.5. Natural mortality

Natural mortality is mortality created by all causes other than fishing, e.g. , predation including cannibalism, diseases, spawning stress, starvation, and old age. Predation and starvation mortality and several others are linked to the ambient ecosystem. The same species may have different natural mortality rates in different areas dependent on the density of predators and competitors whose abundance is influenced by fishing mortality.

There are considerable difficulties in estimating natural mortality. In the worst cases, this has been based on guesswork. For this reason, it is necessary to test the effect of this coefficient term of advice given on the utilization of the stock and in terms of the stock estimate (Stefansson, 1992).

Many authors have treated this problem. Tanaka (1960), Holt (1965) and Saville (1977) demonstrated that the curvature parameter, K , is linked to the longevity of the fish and the longevity is related to natural mortality. Roughly, fish species with a high K - value have a high M - value. Species with a low K - value have a low M - value. Pauly (1980) made a regression analysis of M (per year) on K (per year), L inf. (cm), and T (average annual temperature at the surface in degrees centigrade), based on data from 175 different fish stocks. He based this analysis on the assumption that most of the biological processes occur more quickly at higher temperatures (with certain limits) and, one could imagine that natural mortality was linked to the environmental temperature. Although the above indicated methods are useful tools, the principal methods for the estimation of natural mortality are the analysis on catch-at-age and effort data, and tagging experiments. A lot of research on this subject has been done, yet this is one of the central problems of fisheries research.

In this study, due to the lack of tagging experiments and the high quality data of the effort, it was only possible to estimate the natural mortality by Pauly's method and Tanaka's method.

Tanaka's method relates the natural mortality to the longevity of species, number of survivors at the maximum age and the number of specimens at age of recruitment. This relationship is represented by the following formula:

$$N_i / R = \exp(-M T_m)$$

where M is the instantaneous natural mortality coefficient in units per year, N_i is the number of survivors at age T_i , R is the number of specimens at age of recruitment, and T_m is the lifespan of the species (Tanaka, 1960).

For *Decapterus macarellus*, the instantaneous natural mortality was calculated, assuming the survival rate equally 5% over the lifespan of seven years. The estimated value of M was 0.42 year⁻¹.

Finally, M was estimated by following Pauly's formula:

$$\ln M = -0.0152 - 0.279 \ln L_\infty + 0.6543 \ln K + 0.463 \ln T$$

The growth parameters L_{inf} and K have been estimated from Macdonald's method and $T = 24.1$ is the average surface water temperature computed from the data on average monthly surface water temperature around the Cape Verde Islands available in CRODT (Centre de Recherches Oceanographiques du Dakar - Tiaroy). The estimated M was 0.8. However, Pauly (1983) suggests that the pelagic fishes be accounted for by multiplication by 0.8. Hence, the above estimated value becomes 20% lower.

For further assessments, the M value was taken as 0.4 and maintained as a constant for all ages groups.

5.6 Feeding habits

5.6.1 Stomach contents

242 *Decapterus macarellus* were examined, 100 during the cold season (January - May) and 142 during the warm season (June - December), ranging from 18 to 30 cm FL. Over 90% of the examined *Decapterus macarellus* contained identifiable prey. A total of three prey taxa were analysed. These prey taxa included : crustaceans (copepods, amphipods, euphausiaceans, decapod larvae, and ostracods), mollusks (bivalves, snails), and protozoa (planktonic foraminiferans). The calanoid copepods included the *Aetideidae* spp. and the *Candacia* spp. (Owre et. al., 1967, ; the amphipods included the *Pronoidae* spp. (Brusca, 1980), the euphausiaceans included the *Thysanopoda* spp. (*Thysanopoda tricuspidata*) and the *Euphausia* spp.

(*Euphausia kroni*) (Mauchline, J., 1971), and the snails belong to *Limacina* spp. (Spoel, 1972).

During the cold season (January - May), the most numerous prey groups were copepods (96%), mollusks (1%), and amphipods (1%) (Table 5.6.1.1 and Table 5.6.1.2). Copepods occur most frequently (70%) among the diet items with the molluscs and amphipods occurring in 38% and 15% of stomach respectively. Ostracods and planktonic foraminifera occurred the least frequently in proportions of 3% and 10% respectively.

During the warm season (June - December), the most numerous diet items were decapod larvae (51%), euphausiaceans (21%), copepods (12%), and mollusks (11%). Mollusks, decapod larvae and copepods occur most frequently, in 30%, 20%, and 16% proportions respectively (Table 5.6.1.3).

Table 5.6.1.1 - Long term food diversity (LFD), relative abundance (%N) of prey, frequency of occurrence (%FO), percentage of weight (%W), and the total relative weight of food in stomachs (TRWF) by fish size during the cold season (Jan. - Jun.). N = sample size, A = mean fork length in cm, s = standard deviation of fork lengths, and n = total number of prey.

Length/prey	%N	%FO	%W	TRW	LFD
<20 cm FL: N = 2	A=18	s=1.4	n=1480	1.04	0.41
<i>Aetideidae</i> spp.	82	100	85.23		
<i>Candacia</i> spp.	16	50	11.41		
<i>Pronoidae</i> spp.	2.0	100	3.36		
[20-25] cm FL: N=57	A=23.02	s=1.56	n=11559	0.15	0.96
<i>Aetideidae</i> spp.	84.30	33.33	71.84		
<i>Candacia</i> spp.	13.16	33.33	20.02		
<i>Pronoidae</i> spp.	1.03	24.56	3.99		
Bivalves	0.05	7.02	0.33		
Decapod larvae	0.01	1.75	0.08		
Planktonic foraminifera	0.22	14.04	0.66		
Harpacticoid copepod	0.18	19.30	0.91		
<i>Ostracod</i> spp.	0.01	1.75	0.08		
<i>Limacina</i> spp.	1.05	42.11	2.49		
[26 - 30] cm FL: N=41	A=26.66	s=0.911	n = 3673	0.05	0.97
<i>Aetideidae</i> spp.	93.82	39.02	86.560		
<i>Candacia</i> spp.	4.30	31.71	7.289		
<i>Pronoidae</i> spp.	0.19	4.88	0.456		
Bivalves	0.14	7.32	0.683		
Planctonic Foraminifera	0.14	7.32	0.911		
Harpacticoid copepod	0.03	2.44	0.228		
<i>Ostracod</i> spp.	0.05	4.88	0.456		
<i>Limacina</i> spp.	1.33	34.15	3.417		

Table 5.6.1.2 - Relative abundance of prey (%N), frequency of occurrence (%FO), percentage of weight of prey (%W), and the total relative weight of food in stomachs (TRWF) of the *Decapterus macarellus* at day and at night during the cold season (Jan. - Jun.). Abbreviations as in Table 5.6.1.1

Time/prey	%N	%FO	%W	TRWF
Night	N = 50	A=24, S=1.8	n = 194	0.01
<i>Pronoidae spp.</i>	2.6	4.0	4.55	
Bivalves	4.1	10.0	7.58	
Planktonic foraminifera	14.4	20.0	16.67	
Harpacticoid copepod	10.8	22.0	16.67	
<i>Limacina spp.</i>	68.0	64.0	54.55	
Day	N = 50	A=24.6, s=2.8	n = 16515	0.19
<i>Aetideidae spp.</i>	87.17	72.00	79.490	
<i>Candacia spp.</i>	11.63	66.00	16.802	
<i>Pronoidae spp.</i>	0.91	34.00	2.723	
Bivalves	0.02	4.00	0.116	
Decapod larvae	0.01	2.00	0.058	
Planktonic foraminifera	0.01	2.00	0.058	
Harpacticoid copepod	0.01	2.00	0.058	
<i>Ostracod spp.</i>	0.02	6.00	0.174	
<i>Limacina spp.</i>	0.23	12.00	0.521	

Due to the different sizes of the various prey groups, counting may give a misleading picture of their relative importance in the diet of the capeverdean scad mackerel. In order to get a better idea of the relative importance of the different zooplankton groups, each item was weighed. The results are shown in Tables 5.6.1.1 - 5.6.1.4 and clearly demonstrate that decapod larvae seem to be the most important diet item for the *Decapterus macarellus* ranging from 20 to 25 cm FL if one looks at relative abundance (%N). But the copepods and the euphausiaceans seem to be as important in the diet as the decapod larvae if the analyses are based on percentage of weight values (%W).

Table 5.6.1.3 - Long term food diversity (LFD), relative abundance (%N) of prey, frequency of occurrence (%FO), percentage of weight (%W), and total relative weight of food in the stomachs (TRWF) by size during the warm season (Jul. - Dec.). N = sample size, A = mean fork length in cm, s = standard deviation of fork lengths, and n = total number of prey.

Length/prey	N	A	s	n	TRWF	LFD
<20 cm						
	N = 2	A=19, s=0		n = 18	0.04	0.69
<i>Aetideidae spp.</i>	55.6			50	66.67	
Decapod larvae	0.44			50	33.33	
[20 - 25] cm						
	N = 140	A=212, s=1.1		n = 2171	0.03	1.30
<i>Aetideidae spp.</i>	12.2			16.4	10.10	
<i>Candacia spp.</i>	0.4			1.4	0.38	
<i>Pronoidae spp.</i>	0.4			5.0	1.35	
Bivalves	11.8			30.0	9.35	
Decapod larvae	51.2			20.7	36.64	
Foraminifera	0.5			3.6	0.96	
Harpacticoid copepod	0.05			0.7	0.19	
<i>Euphausiacea spp.</i>	21.1			17.9	39.87	
<i>Ostracod spp.</i>	0.1			2.1	0.192	
<i>Limacina spp.</i>	0.3			2.1	0.58	

On the whole, during the cold season, copepods constituted 95%, by weight, of the food taken by the capeverdean scad mackerel, while the other items make up the remaining 5% left, and during the warm season the copepods constitute 15%, the decapod larvae 30% and the euphausiaceans 30%.

Some differences were found in stomach contents of fish of different sizes (Tables 5.6.1.1 and 5.6.1.3). Small fish preyed almost exclusively on copepods (although very few fish are included in the analyses). Medium-sized fish preyed predominantly on copepods, but to a lesser extent than small fish. Large fish fed on a variety of prey and consumed large prey such as euphausiaceans. The long term food diversity (LFD) shown in the Tables and which describes the food diversity potential of the species, varies from 0.41 for small fish to 1.3 for large fish which confirms that the large fish are more diversified in their feeding.

Table 5.6.1.4 - Relative abundance of prey (%N), frequency of occurrence (UFO), percentage of weight of prey (%W), and the total relative weight of food in stomachs (TRWF) of the *Decapterus macarellus* by early morning (24 - 09), middle of day (10 - 16), and later day (18 - 23) during the warm season (Jul. - Dec.). Abbreviations as in Table 5.11.1.

Time/prey	%N	%UFO	%W	TRWF
[24 : 09] hours				
	N = 30	A=21.1, s=1.1	n = 5	0.0005
<i>Candacia spp.</i>	100	3.3	100.00	
[10 : 16] hours				
	N = 61	A=21, s=0.7	n = 1849	0.1
<i>Aetideidae spp.</i>	12.7	37.7	10.21	
<i>Candacia spp.</i>	0.2	1.6	0.22	
<i>Pronoidae spp.</i>	0.4	11.5	1.51	
Bivalves	0.8	6.6	0.86	
Decapod larvae	60.5	49.2	41.15	
Foraminifera	0.4	6.6	0.86	
<i>Euphausiacea spp.</i>	24.8	41.0	44.55	
<i>Ostracod spp.</i>	0.2	4.9	0.65	
[18 : 23] hours				
	N = 52	A=21.6, s=1.2	n = 341	0.01
<i>Aetideidae spp.</i>	11.7	1.9	24.7	
Bivalves	71.0	73.1	78.80	
Foraminifera	0.6	3.8	1.77	
Harpacticoid copepod	0.3	1.9	1.77	
<i>Limacina spp.</i>	1.8	5.8	5.30	

Some differences were found in the stomachs of the fish collected during different times of the day (see tables above). The capeverdean scad mackerel fed almost exclusively during the daylight hours during cold season. During the warm season, the capeverdean scad mackerel shows the same pattern in his feeding time as in the cold season, feeding almost exclusively during the time period 10:00 hours to 16:00 hours. If one takes a closer look at the total relative weight of food in the stomachs (TRWF) collected during different times of day that are shown in the tables above, it should not be difficult to see that the TRWF values for the period 10:00 to 16:00 hours are tenfold over than those for other time periods. Relatively, this means that capeverdean scad mackerel, during the daylight hours, feed ten times more than during the night time.

5.6.2 Discussion

Zooplanktivores feed during the day or night, but seldom during both periods (Hales, 1987). The carangids like *Decapterus macarellus* are diurnally active, and frequently form feeding flurries right at the surface that are sufficiently prominent to attract fishermen and fish-eating birds from considerable distances. As the twilight arrives, the schools disperse and migrate over the sand where they forage on benthic and midwater organisms during the night (Longhurst, A. and Pauly, D., 1987 and personal observation).

The strongly diurnal periodicity of stomach fullness indicates that capeverdean scad mackerel feed mostly during the daylight hours. The presence of some food in the stomachs of the capeverdean scad mackerel caught during the night hours suggests that some nocturnal feeding may be occurring.

Scales were a common item found in the capeverdean scad mackerel stomachs, but were discredited because several observations (personal observations) indicated that the capeverdean scad mackerel were feeding on scales abraded by fish during their avoidance of, or capture by purse seine. Our study thus shows no sign of cannibalism.

Ontogenetic changes in the diet occurred with growth. Larger fish had more diverse diets which included larger prey (euphausiaceans).

Seasonal changes in the diet of the capeverdean scad probably reflect fish size and the relative abundance of zooplankton components (holoplankton and meroplankton). Copepods, decapod larvae, euphausiueceans and bivalves were the most numerous. The relatively high frequency of molluscs (bivalves and *Limacina* spp.) in the stomachs of the capeverdean scad mackerel, either during the day or during the night, may be attributed to both the abundance of molluscs in the water off the Cape Verde Islands, and to their low digestibility and long retention in stomachs (Donaldson *et. al.*, 1994, Hales *et. al.*, 1987)

One could not conclude that the seasonal changes in the diet of the capeverdean scad mackerel are only due to the seasonal changes in the abundance of zooplankton in the water off the Cape Verde Islands, since nothing is known of the temporal pattern of productivity of phytoplankton and zooplankton in the area.

6. ASSESSMENT OF SCAD MACKEREL (*Decapterus macarellus*, Cuvier 1833) STOCK

6.1 ASSESSMENT THEORY

6.1.1. Estimation procedure

Usually, most stock estimations are based on the classical Virtual Population Analysis (VPA) of Gulland (1969). However several variations exist on VPA. A simple variation is the so-called Cohort Analysis which assumes that fishing takes place around the middle of the year and that natural mortality will affect the stock before and after the fishing season. Since the Age Cohort Analysis, the Length Cohort and Tuning Cohort Analysis have been used in estimation of scad mackerel stock size, so they will be described below with more emphasis.

The cohort analysis was developed by Pope (1972) and reviewed by Gulland (1983) and Jones (1981).

Pope's cohort analysis is based on the assumption that the catch from the cohort is taken instantaneously on the first of July, consequently, during the first half of the year the fish suffer only natural mortality. Thus the number of survivals of a cohort is described mathematically by the following equation:

$$N_{a+1,y+1} = \left[N_{ay} e^{-\frac{M}{2}} - C_{ay} \right] e^{-\frac{M}{2}} \quad (6.1.1.1)$$

or

$$N_{ay} = \left[N_{a+1,y+1} e^{\frac{M}{2}} + C_{ay} \right] e^{\frac{M}{2}} \quad (6.1.1.2)$$

The calculations are much easier than in VPA analyses because here "F" does not occur. But the calculations start the same way as in the VPA by assuming that "F" for the oldest age groups are an average of fishing mortality rates for younger age "o groups" and that mortality coefficients for the last year are given.

From the estimate of N's, the estimate of fishing mortality is obtained from:

$$F_{ay} = \ln \frac{N_{ay}}{N_{a+1,y+1}} - M \quad (6.1.1.3)$$

From the following equation it is possible to calculate the stock size for each age-group in each year:

$$N_{t+1,y+1} = N_{t,y} e^{-F_{t,y} \Delta t} \quad (6.1.1.4)$$

If we consider the catch during any time period, so the equations (6.1.1.2) and (6.1.1.3) must be written in the following forms:

$$N_t = [N_{t-1,y} e^{(M/2)\Delta t} + C_{t-1,y}] e^{(M/2)\Delta t} \quad (6.1.1.5)$$

$$F_{t,y+1,y} = \frac{1}{\Delta t} \ln \frac{N_t}{N_{t-1,y}} - M \quad (6.1.1.6)$$

The differences between VPA results and cohort results are very small. But the calculations with the cohort analysis are much easier.

In a situation where only length composition data for the total fishery is available (or average length for a sequence of years), we should work with the called "length based cohort analysis". Here the real cohort is replaced by the "pseudo-cohort by assuming the constant parameter system, i.e. assuming that the class ages in one year reflect that of the cohort during its life-span.

To convert the cohort analysis equation (6.1.1.5) into a length based version only the term with "exp." has to be changed. So the equation (6.1.1.5) will become in the following:

$$N(L1) = \left[N(L2) \left(\frac{L_{\infty} - L1}{L_{\infty} - L2} \right)^{(M/2K)} + C(L1, L2) \right] \left(\frac{L_{\infty} - L1}{L_{\infty} - L2} \right)^{(M/2K)} \quad (6.1.1.7)$$

and the equation (7.2.1.6) transform in follow form:

$$F = M \frac{F/Z}{1 - F/Z} \quad (6.1.1.8)$$

where F/Z is:

$$F/Z = \frac{C(L1, L2)}{N(L1) - N(L2)} \quad (6.1.1.9)$$

Here, in the length converted cohort analysis, it is necessary to make an initial guess at the exploitation rate, F/Z , rather than the terminal F . Usually, it is assumed that the exploitation rate for the last length group has a value of 0.5 (Sparre and Venema, 1992).

After all the procedures of the cohort analysis shown above (we estimate some F for last year) are carried out, we need some additional data for the tuning cohort analysis. These data are the effort for each year, the time series of the survey index, and the CPUE.

From the cohort analysis and weight estimates, we calculate the total biomass and fishable biomass from the number at age and an average F . Then we correlate the three series with the additional data: Average F versus effort, total biomass versus survey index and fishable biomass versus CPUE. By assuming different values of F at the oldest age for the cohort analysis, then different values of average of F , total biomass and fishable biomass can be computed. By comparing the correlation coefficients we can determine which " F " at the oldest age is best for the cohort analysis.

6.2 Stock estimates

In this study, an assessment was conducted on the capeverdean scad mackerel, using catch-at-length and catch-at-age data from 1983 - 1995 (Table 5, table 8), and catch in weight and effort data from 1986 - 1992 (Table 1).

In this assessment, a natural mortality of 0.4 per year was used for all ages and length groups (see chapter 5.5).

Originally, a worksheet in Excel was prepared based on the age cohort analysis formulas (chapter 6.1) and tuned with the standardized effort data. This means that a least squares method was used to minimize the sum of squared errors between the observed standardized effort and the effort estimated. The estimated effort per year was calculated from the average fishing mortality rate in each year and then multiplied by the catchability (q). An interactive process was run in order to establish one reasonable terminal fishing mortality.

However, due to the great variability in observed effort data, a reasonable terminal fishing mortality was not reached. So the tuning was left out, and the length's cohort (Jones, 1981, and Sparre and Venema, 1992) was run, using the catch-at-length data and the growth parameters found in chapter 5.1 in order to estimate a reasonable terminal fishing mortality. Table 7 in the appendix shows the results of the length's cohort. An average fishing mortality of $F = 0.15$ was considered reasonable and taken as an educated guess of the terminal fishing mortality in the

age cohort analysis. A selection pattern in the last year was assumed, based on the average selection pattern computed in the period 1983 - 1994. Hence, an age cohort analysis was run, using the catch-at-age data, and the stock size, fishing mortalities, and selection patterns for each age group and for each year were computed and are available in Tables 8 - 13.

As we see in some age groups the fishing mortality reaches values of 3 or even 5. Due to inadequate sampling (oversampling in some age-groups and undersampling in others), the sampling bias was carried over into the computed catch-at-age and hence into the size at age and fishing mortalities at age. But, if we look carefully at the table of fishing mortality coefficients one could say without reserve that the coefficients typically range from 0.05 to 0.7 per year.

By analyzing the scad mackerel stock size in the period from 1983 - 1995 (see Table 9 in appendix), we see strong years in recruitment to the fishery in 1984, 1986, 1989, and 1991. To explain the appearance of these strong year classes is a difficult task. The cause could be environmental changes in the waters off the Cape Verde Islands such as availability of food (copepods and decapod larvae, see chapter 5.6), but this is difficult to prove due to lack of information on plankton composition in the water during this period (see chapter 5.6). The author does not believe that the appearance of those strong year classes is due only to the biased sampling that carried over into the stock size computations.

6.3 Estimation of spawning stock and recruitment

The spawning stock is the quantity of fish that spawn for each year. This quantity is, therefore, measured in tons. The size of the spawning stock in the spawning season is computed by taking the mass number for the beginning of the year and making allowances for mortality until the onset of the spawning season (Stefánsson, 1992). The corresponding year's class sizes are then multiplied by the ratio of mature fish in each age group and the mean weight of each age groups.

Recruitment is simply defined as the number of fish in the youngest year-class. Generally, the number of youngest fish in the sea that are included in the catch is used as a means to assess the recruitment.

For the capeverdean scad mackerel, like other tropical fish which spawn throughout the year (although the capeverdean scad mackerel spawns from March to August),

the spawning stock was calculated by taking the stock size corresponding to the whole year's class sizes and multiplying by the maturing ratio in each age group computed in chapter 5.4 and the corresponding mean weight at age computed in chapter 5.1.

Since scad mackerel that belong to age-group 1 (mean length equal to 9 cm FL) does not recruit to the fishery (at least a scad mackerel with the fork length under 15 cm is hardly seen in the landing), the recruitment was assumed to be equal to the estimated number of fish that belong to age-group 2.

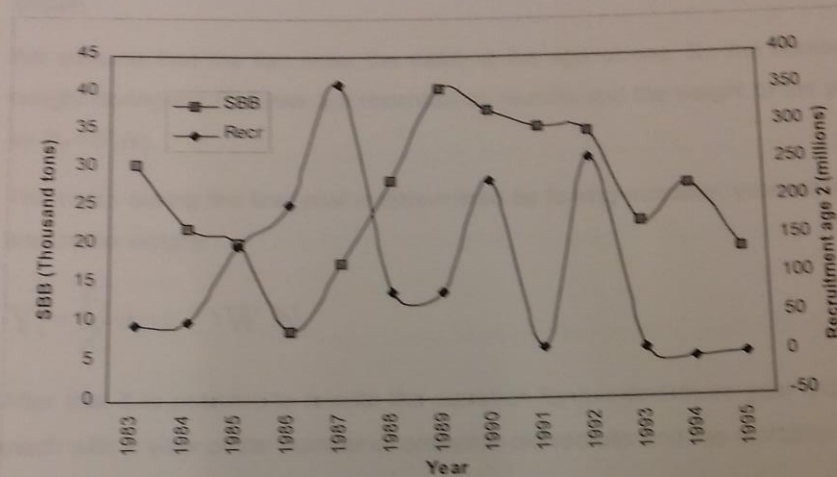


Fig. 6.3.1. - The spawning stock and the recruiting year class (2-group) by number of capeverdean scad mackerel in the years 1983-1995.

As is shown in Fig. 6.3.1, recruitment fluctuates throughout the period from 1983-1995 with high values in 1987, 1990, and 1992 and lower values in 1983, 1984, 1991, 1994, and 1995. These anomalous values are explained by undersampling during the recruitment seasons mainly in the years 1994 and 1995 (see chapter 5.1). The average recruitment in the above indicated period is about 180 millions. The author does not believe that the fluctuations in the recruitment level are due to overfishing.

The spawning stock fluctuates less throughout the period from 1983-1995. The only anomalous value is seen in 1986 with an SSB equal to 9 thousands tones, but if we take a look at the length-frequency data, we obviously will see that in 1986 the sampling was done mainly on specimens with an FL less than 25 cm which have a lower maturing ratio. The sampling took place during a recruitment season when the

proportion of maturing scad mackerel in the commercial catches was small. So, in 1986 there was a clear undersampling of the mature portion of capeverdean scad mackerel.

6.4 Yield per recruit and Spawning stock per recruit

In the fish stock assessment, the potential catch per recruit is determined in order to examine how much can be caught of a specific year class when its size by number is known.

We assume that the fish enter the catch at the age of one. So the number of fish caught during the first year are recorded as recruits and the weight of the year class as $B_1 = W_1 N_1$.

The catch during the first year is determined by fishing mortality, year-class strength and mean weight:

$$Y_1 = \frac{F_1}{Z_1} (1 - e^{-Z_1}) W_1 N_1 \quad (6.4.1)$$

After that it is possible to rewrite the equation for two and more year older fish in a catch with a view of the number of one-year-old recruits and the mortality coefficients between them.

The catch at age is given by:

$$Y_x = \frac{F_x}{Z_x} (1 - e^{-Z_x}) W_x e^{-\sum_{i=1}^{x-1} Z_i} R \quad (6.4.2)$$

And the yield per recruit is arrived by:

$$\frac{Y}{R} = \sum_x \frac{F_x}{Z_x} (1 - e^{-Z_x}) W_x e^{-\sum_{i=1}^{x-1} Z_i} \quad (6.4.3)$$

It is possible to compute the spawning stock per recruit in the same way as we calculated the yield per recruit (Stefánsson, 1994).

The yield per recruit model gives valuable indications as to whether a heavy or light effort (or fishing mortality) is more feasible and whether it is better to catch young or old fish.

Graphics showing yield per recruit always increase initially with increasing F , showing zero yield when there is no effort and increasing up to a point when additional effort provides negligible additional catch. Sometimes, yield per recruit then decrease when the effort exceeds a certain limit. Sometimes, the yield per recruit continues to rise, but gradually less and less. It is therefore always desirable to plot the yield per recruit and the spawning stock per recruit against fishing mortality, because then it will be much easier to do some important analyses (like determining the Maximum sustainable yield per recruit).

Here, computations of yield per recruit and spawning per recruit for the capeverdean scad mackerel were carried out by using a spreadsheet made in Excel. The input quantities for the above computations are shown in Table 6.4.1.

The weight-at-age, the proportion mature in the stock, and the selection pattern used here are based on the average weights-at-age, and the maturity, and the selection pattern for the period 1983-1995, discussed previously in the sections 5.4, 5.7, and 6.3 respectively.

The yield per recruit and the spawning stock per recruit plots against fishing mortality are given in Fig 6.4.1.

Table 6.4.1.- Major input quantities for the computations of yield per recruit and spawning stock per recruit for the capeverdean scad mackerel

Age	2	3	4	5	6	7
Natural mortality coefficient	0.4	0.4	0.4	0.4	0.4	0.4
Proportion mature	0.113	0.355	1.0	1.0	1.0	1.0
Mean weight in catch (kg)	0.042	0.076	0.130	0.160	0.218	0.253
Fishing pattern	0.28	1.0	1.0	1.0	1.0	1.0

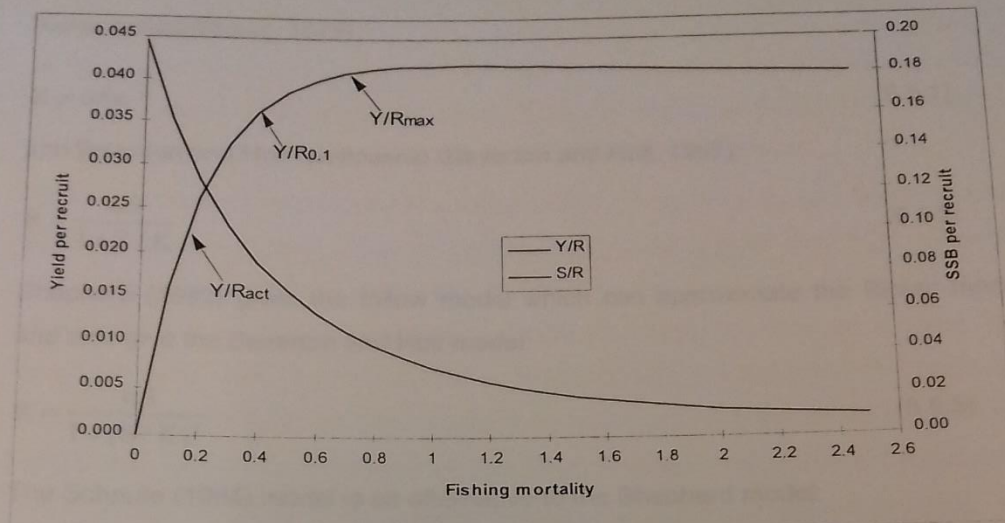


Fig 6.4.1 - Yield per recruit (Kg/recruit) and spawning stock per recruit (Kg/recruit) of the capeverdean scad mackerel.

From the analysis of the yield per recruit plot, the F_{max} , which provides maximum yield per recruit of 41.3 grams, is equal to 0.8 per year. The maximum spawning stock biomass per recruit is equal to 188 grams. The so-called $F_{0.1}$, which provides the maximum sustainable yield per recruit of 40 grams, is equal to 0.4 per year.

It is noted here that, by assuming an average recruitment of 180 millions (see chapter 6.3), one would expect an average yield of 7.5 thousand tons at $F=0.8$, and the Y/R is 41.3 grams. Furthermore, there is much incentive to reduce the fishing mortality from the level of 0.8 to $F_{0.1}=0.4$, since the biomass per recruit will change from 43.6 grams per recruit to 84.1 grams per recruit and one could expect the average yield to decrease only in by 10%.

However, care should be taken when a method like this is used to obtain information concerning the yield because this method is based on long term data and usually it is wise and safe to assume that various conditions have changed during the study period (mean weight at age, maturing ratio, fishing pattern, etc.).

6.5 Spawning stock and Recruitment

Several models exist for describing the relationship between spawning stock biomass and recruitment. The most common are the following:

Ricker model (Ricker, 1975) :

$$R = \alpha S e^{-\frac{S}{K}} \quad (6.5.1)$$

and Beverton and Holt relationship (Beverton and Holt, 1957):

$$R = \frac{\alpha S}{1 + S / K} \quad (6.5.2)$$

Shepherd (1982) gives the follow model which can approximate the Ricker model and also give the Beverton and Holt model:

$$R = \frac{\alpha S}{1 + (S / K)^\gamma} \quad (6.5.3)$$

The Schnute (1984) model is an alternative to the Shepherd model:

$$R = \frac{\alpha S}{(1 + S / \gamma K)^\gamma} \quad (6.5.4)$$

$$\text{Cushing} \quad R = \alpha S^\beta \quad (\gamma=0) \quad (\text{Jakobsson et. al., 1993}) \quad (6.5.5)$$

When $\gamma \rightarrow 1$ this model reduces to that of Beverton and Holt and to Ricker's model when $\gamma \rightarrow \infty$.

Essentially, the Beverton and Holt model says that, above a certain level (αK) of spawning stock, there is no relationship between the parent stock and the recruitment, whereas the Ricker model says that this relationship exists for all sizes of spawning stock, and there is an optimum spawning stock size.

In the Beverton and Holt model, "K" is the size of spawning stock that produces half the maximum recruitment. In the Ricker model, maximum recruitment is obtained when the spawning stock is "K".

Data on spawning stock biomass (SSB) and recruitment at age 2 are given in Table 9 and Table 11 in appendix. These data have been analyzed to investigate what kind of stock-recruitment relationship best fits the data. The stock-recruitment relationships investigated included the Ricker model, the Cushing model, and the Beverton/Holt model which are well described above.

These models have been fitted by minimizing the residual sum of squares from each stock-recruitment relationship log-transformed (Stefánsson, 1994).

The fitted models by the least squares methods have provided unreasonable results. As Shepherd (1982) points out, poor fits may be explained by the poor quality of the data or obscured by the effects of a fluctuating environment, rather than by a lack of relationship between spawning stock and recruitment does not exist.

Hence, a method offered by Shepherd (1982) is used to fit the Beverton/Holt stock/recruitment function. First we should examine the ratio of R to SBB (for each year). Then the inverse of the median of this ratio denotes the typical production of the spawning stock. The corresponding mortality is called F median. In our case, the computed slope is equal to 11 and the corresponding F_{med} is equal to 0.3. The above computed slope is used as a starting point for α . By guessing values for α and β , we ensure that values $\alpha \times \beta$ will be not more than the maximum observed recruitment and that the selected curve passes through the observed spawning stock and recruitment values. This will yield results consistent with the current state of the fishery, and we will then be able to find reasonable estimates for α and β . The Beverton/Holt curve of $\alpha = 19$ and $\beta = 18$, generated in this way for the spawning stock and recruitment relationship of the capeverdean scad mackerel, is illustrated in Fig. 6.5.1.

Although the method could be subject to scientific judgment, however, this should be considered to be an advantage, rather than a disadvantage, since formal statistical methods of parameter fitting on scattered data like our case can very easily yield ridiculous results. This has happened when we have done fitting using the least squares method.

Fig. 6.5.1 shows us the best fit (Bev/holt function) for the stock/recruitment relationship for the capeverdean scad mackerel based on data in the period from 1983 - 1995.

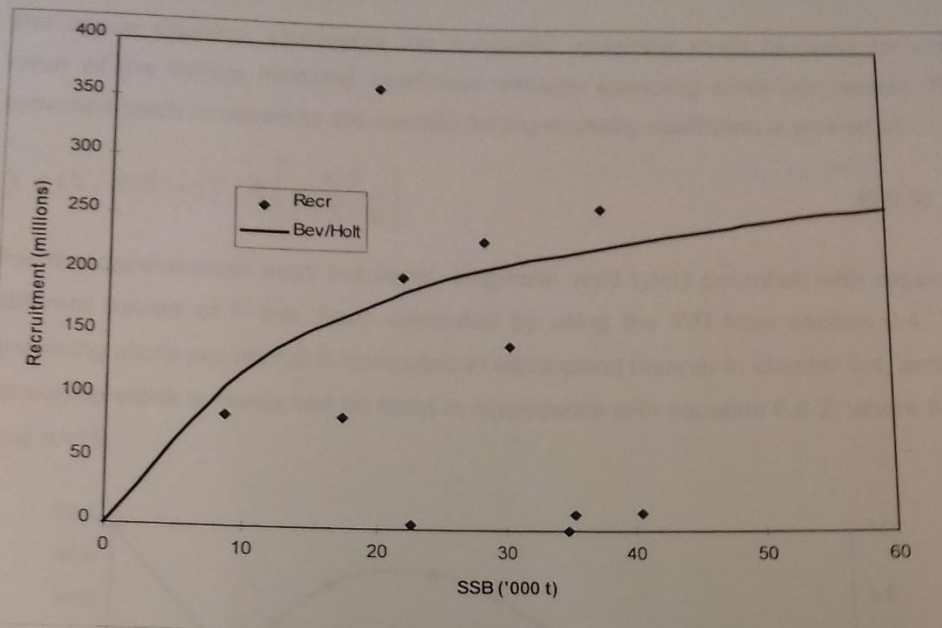


Fig 6.5.1 - Spawning stock biomass (thousand tons) and recruitment (millions) along with "fitted" Beverton/Holt relationship.

6.6 Yield Potential

When computations for yield per recruit, spawning stock per recruit and the relationship between recruitment and spawning stock have been completed, it is possible to compute the potential yield of the stock with regard to the different fishing mortalities.

By assuming that yield per recruit (Y/R) and spawning stock per recruit (S/R) are known for each value of fishing mortality (F) and that the stock/recruitment relationship has been assessed by the Beverton/Holt form, then information can be linked together in a relatively simple fashion. From the Beverton/Holt equation, the spawning stock could be written as :

$$S = (S/R) \frac{\alpha \cdot S}{1 + S/K} \quad (6.6.1)$$

By using the above equation, the spawning stock is denoted by:

$$S = K[\alpha \cdot (S/R) - 1] \quad (6.6.2)$$

The above equation represents the computed spawning stock biomass for each value of the fishing mortality coefficient through spawning stock per recruit. The expected catch in regard to the specific fishing mortality coefficient is arrived at :

$$Y = (Y/R)R = (Y/R) \left[\frac{\alpha \cdot S}{1 + S/K} \right] \quad (6.6.3)$$

For the capeverdean scad mackerel, long-term yield (yield potential) with regard to different values of F has been computed by using the Y/R from section 6.4. The spawning stock per recruit is computed in correspond manner in chapter 6.4, and the spawning stock is computed (in tons) in accordance with equation 6.6.2, where $K=18$ and $\alpha=19$.

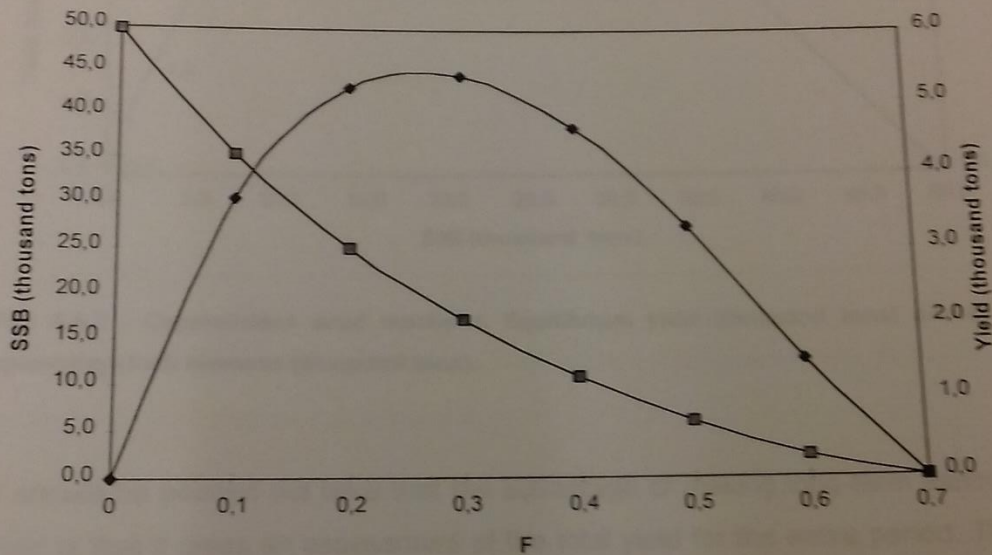


Fig 6.6.1 - Long-term sustainable yield and spawning stock biomass against fishing mortality for the capeverdean scad mackerel fishery.

The long term sustainable yield for different values of F are illustrated in Fig. 6.6.1. The maximum sustainable yield reached at a fishing mortality equal to 0.3 is about 5.5 thousand tons. It is seen here that with an increasing value of F there is considerably more yield lost when the stock recruitment relationship is taken into account, than would seem to be the case judging from the Y/R results in Fig. 6.4.1 (see chapter 6.4). It should be noted here that the estimation of F_{max} is reduced on a large scale (from 0.8 to 0.3).

Since the current catch is around 3 thousand tons (1996) corresponding to a fishing mortality of 0.15 per year, and the maximum sustainable yield is about 5.5 thousand tons, then one could say that there is room for increasing the total catches from a level 3 to 5 - 6 thousand tons without putting the stock at risk (see Figure 6.6.2).

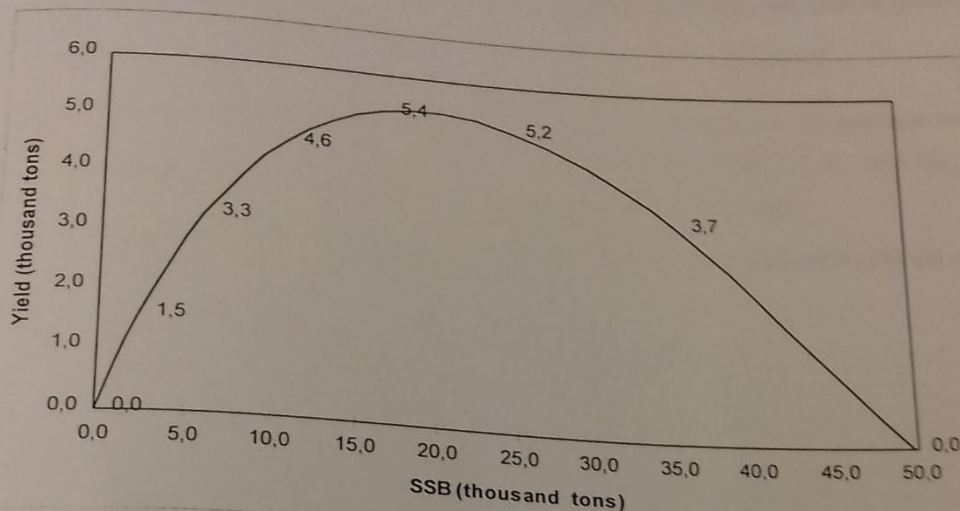


Fig. 6.6.2 - Capeverdean scad mackerel. Equilibrium yield (thousand tons) plotted against spawning stock biomass (thousand tons).

It should be pointed out here that the advantage of making long-term estimations of yield is that it gives an assessment of the total yield for the entire period. Thus, it is possible to visualize the long term gain of constructive measures in relation to the short-term loss of a temporary decrease in fishing mortality (Stefánsson, 1994). However, there are risks, due mainly to the fluctuations of the stock sizes. The spawning could oscillate around a given value and fail to reach an equilibrium. It must be noted here that the above indicated method for computing long-term yield is based on an equilibrium system.

Of course, to the model simulators for the effects of increasing or decreasing fishing mortality from a fixed point could be easily add or simulation of certain variations in recruitment carried out according to the Beverton/Holt relationship (Baldursson et al., 1996). This is further treated in the next chapter.

6.7 Catch Prediction and risk analysis

The outcome of the stock estimations usually states the number of fish in the sea at the beginning of the last year for which the data is available. Herein, the capeverdean scad mackerel will enter the stock at age 2 and will reach age 7 at the most. The number of fish in the cohort at the beginning of the next year is given by:

$$N_{a+1,y+1} = N_{a,y} \exp(-s_{a,y} F_y - M_{a,y}) \quad 2 \leq a \leq 7$$

where F_y is the overall fishing mortality in the year Y (defined as the average over the ages 3 - 6), $s_{a,y}$ is the fishing pattern (estimated in the section 6.2), and $M_{a,y}$ is the natural mortality which is assumed to be equal to 0.4 for all ages.

Hence, the number of fish caught, according to the Baronov equation (1918) is:

$$C_{a,y} = \frac{s_{a,y} F_{a,y}}{(s_{a,y} F_{a,y} + M_{a,y})} (1 - \exp(-s_{a,y} F_{a,y} - M_{a,y})) N_{a,y}$$

and the catch in weight during year y is:

$$Y_y = \sum_{a=2}^7 w_{a,y} C_{a,y}$$

where the $w_{a,y}$ is the average weight of fish in the catch of each age group in the year y . The exploited biomass is defined as:

$$B_y = \sum_{a=2}^7 w_{a,y} s_{a,y} N_{a,y}$$

The spawning stock is computed as:

$$S_y = \sum_{a=2}^7 w_{a,y} p_{a,y} N_{a,y} \exp(-s_{a,y} F_y - M_{a,y})$$

where $p_{a,y}$ is the fraction of fish of reproductive capacity in each age group in year y .

Then a Beverton/Holt recruitment model is used to relate the age-group 2 at the beginning of the year, to the size of the spawning stock two years earlier (see chapter 6.5).

The long-term values of parameters, used in catch prediction simulations, are presented in Table 6.7.1.

Table 6.7.1. Input data simulations. Starting values and long term values for proportion mature at age-groups (p), weight at age-groups (w), and selection pattern (s).

Age	p	w (grams)	s	N ₁₉₉₅ (millions)
2	0.01	42	0.28	187.87
3	0.36	76	1	0.014
4	0.75	130	1	4.80
5	1	160	1	76.97
6	1	218	1	0.011
7	1	253	1	23.86

The proportion mature, weight-at-age, and selection were averages of the estimated values for the years 1983 - 1995. The initial stock sizes⁶ are estimated in chapter 6.2 and are shown in Table 6.7.1. The long-term values were used from 1995 to 2005. The values of parameters in the recruitment function were $\alpha=19$, $K=18$. The function gives the prediction of recruitment in terms of millions.

The overall fishing mortality in the starting year (1995) used here is equal to 0.15 which has been computed in chapter 6.2.

In order to make predictions for catches and spawning stock biomass until 2005, random fluctuations in recruitment and in F_y have been generated by multiplying the recruitment function by a log-normal variable of the form $\exp(1.016X - 1.016^2)$, where X is random variable and 1.0 is the standard deviation of log-recruitment values estimated by age cohort analysis in the period 1983 - 1995. We also multiply $F=0.15$ by a log-normal variable of the form $\exp(0.6X - 0.6^2)$, where X is a random variable and 0.6 is the standard deviation of log-average fishing mortality ($F_{1987-1993}$). Each simulation will thus provide us with examples of possible developments of spawning stock biomass and catch for a few years into the future.

About 100 simulations have been carried out in order to investigate what kind of development is likely and unlikely to occur in the catch and spawning stock biomass of the capeverdean scad mackerel stock until the year 2005. The major results found from these simulations are illustrated in Figs. 6.7.1 - . 6.7.3.

⁶ The size for age-group 2 has been computed using the Beverton & Holt recruitment function with $\alpha=19$ and $k=18$.

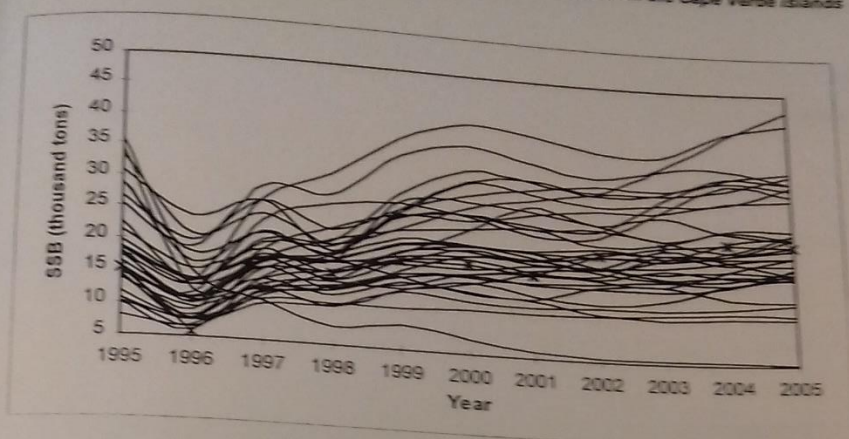


Fig. 6.7.1- Spawning stock of the capeverdean scad mackerel: 100 simulation runs

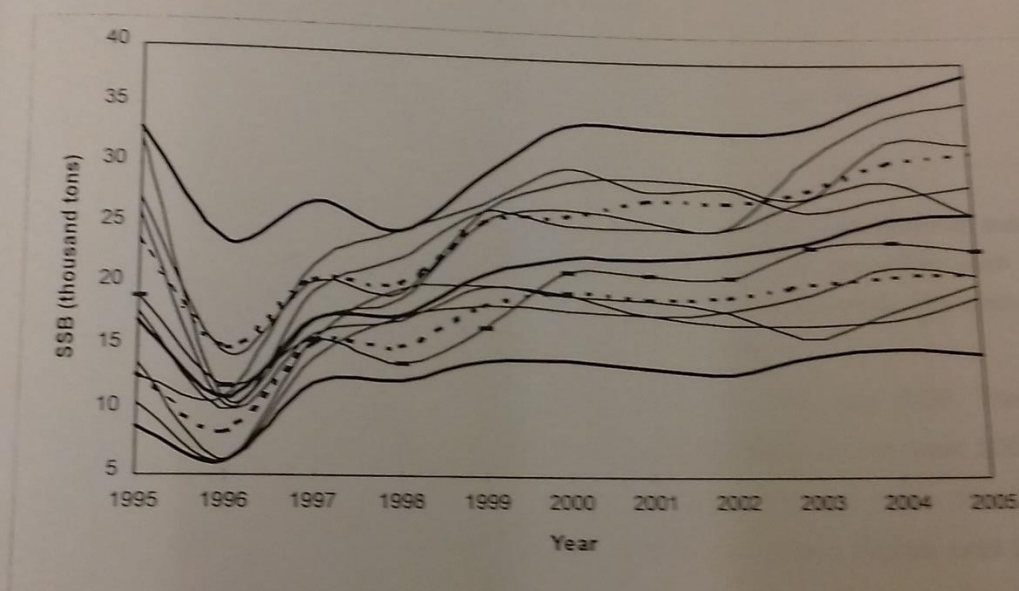


Fig. 6.7.2- Catch development based on 100 simulations. Top and bottom, bold, solid curves: 95% and 5% percentiles in each year, respectively. Central, bold, solid curve: median for each year. Dashed curves: 25% (below) and 75% percentiles (above). Thin lines: sample simulated trajectories.

Figure 6.7.1 shows 100 simulation paths for the spawning stock of the capeverdean scad mackerel. As can be seen from Figure 6.7.1, the stock collapses in very few cases.

Figure 6.7.3 shows several bold lines which denote percentiles for the spawning stock each year. Specifically, the probability is 5%, 25%, 50%, 75%, and 95% that the spawning stock biomass of the capeverdean will be below the corresponding line in a given year. To show fluctuations from year to year, the figure also shows five

time paths from simulations. The median size of the spawning stock will be around 24 thousand tons in the year 2000 and the stock will eventually stabilize at around 25 thousand tons.

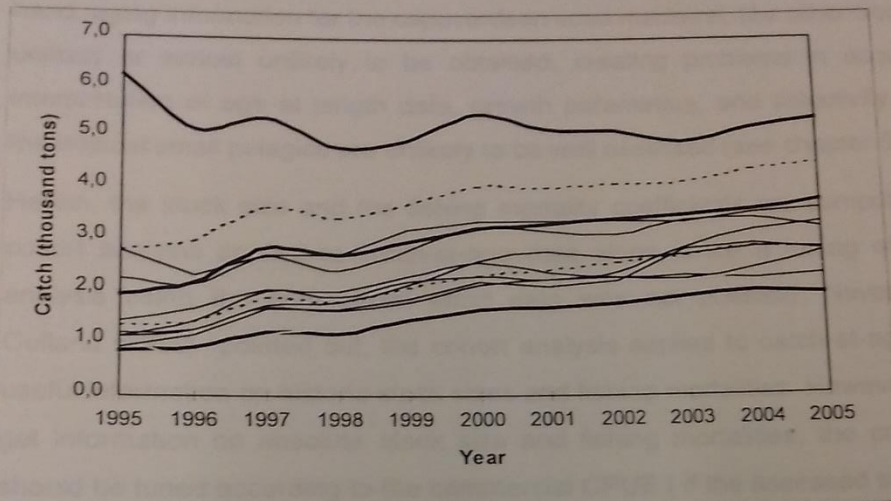


Fig. 6.7.3 Spawning stock biomass development of the capeverdean scad mackerel based on 100 simulations. Top and bottom, bold solid curves: 95% and 5% percentiles each year, respectively. Central bold, solid curve: median for each year. Dashed curves: 25% (below) and 75% percentiles (above). Thin lines: sample simulated trajectories.

Figure 6.7.2 shows the development of the catches until the year 2005. The catch reaches 3.5 thousand tons at the 50th percentile (median) by the year 2000, and then continues to increase slightly. At the 95th percentile the catch reaches 4.2 thousand tons by the year 2000 and then continues to increase slightly until the year 2005.

It should be pointed out here that no management procedure has been used to simulate the catch and spawning stock development. Neither the minimum catch nor the maximum catch have been taken into account in the simulations. A management procedure which could be dependent on social, economic and political reasons could either minimize the probability of a collapse of the stock or maximize the probability of a collapse of the stock. This is outside of the scope of this study. However, it would be interesting to see how the management procedure could affect the stock development of the capeverdean scad mackerel in the future.

6.8 Discussion

Decapterus macarellus, like other small shoaling pelagic fish stocks, usually does not fit well into the traditional population dynamic models and assumptions, thus making their assessment and management particularly difficult and uncertain. On the other hand, aging information for the capeverdean scad mackerel, like other tropical fish, is unlikely or almost unlikely to be obtained, creating problems in acquisition and interpretation of age at length data, growth parameters, and selectivity. Therefore, the tropical small pelagics are unlikely to be well assessed (see chapter 5.1).

Herein, the stock size and the fishing mortality coefficients are computed from the cohort analysis applied to catch-at-age data alone, since a tuning of age cohort analysis using the commercial effort data was not possible. Nevertheless, like Gulland (1988) pointed out, the cohort analysis applied to catch-at-age data gives useful information on historic stock sizes and fishing mortalities. However, in order to get information on absolute stock size and fishing mortalities, the cohort analysis should be tuned according to the commercial CPUE (if the assessed species is non-shoaling), or according to the estimates of stock sizes provided by acoustic surveys (for shoaling species).

However, due to the lack of time series of acoustic surveys on capeverdean scad mackerel, the whole assessment should rely on data that is available.

All of the procedures and methods used herein contain uncertainty both in specific parameter estimates and the resulting management advice. For example, the age cohort analysis used here relied on the use of a numerical conversion method (Macdonald method) to estimate the age composition. The resultant uncertainty from the numerical conversion method is carried over to the age cohort analysis.

One of the most difficult problems found here in this assessment was estimating the stock recruitment relationship. The Shepherd method (1982) was used to quantify the stock recruitment relationship for the capeverdean scad mackerel (chapter 6.5), and this resulted in uncertainty in setting biological reference points and in determining the potential yield. However, the stock recruitment parameters, the biological reference points and the potential yield computed seemed to be consistent with the reality of the fishery.

In order to provide a more realistic setting for the provision of short-term management advice for the capeverdean scad mackerel, a risk analysis has been

applied to the catches and the SSB of the capeverdean scad mackerel. As is shown in chapter 6.6, there are no signs that the stock could collapse until 2005.

A "prudent exploitation" strategy therefore seems to be the best approach for reducing the potential risk of collapse and this requires that the maximum level of fishing be set below the estimated level required to achieve the maximum equilibrium yield. The fishery should also be carefully monitored. Management should be prepared to take adequate action (i. e. early and strong enough) at the first sign of fish stock deterioration and an increased risk of collapse.

7. CONCLUSIONS AND RECOMENDATIONS

The capeverdean scad mackerel fishery seems to be one of the most important fisheries at Cape Verde and it deserves more attention. This requires careful monitoring starting with a more adequate sampling program and wise management.

One of the most important conclusions, and the most pertinent, is that there is a huge need for improving data collection on the length frequency. It is seen here that all of the biological parameters estimated, and hence the whole assessment relied exclusively on the quality of the length-frequency data, since age reading has been proved to be unreliable. A more random sampling of length frequency should be carried out in order to make improvements.

Analysis of feeding habits shows that the capeverdean scad mackerel feeds mainly during the daylight hours near the surface, preying almost exclusively on copepods and decapod larvae. However, the diet and feeding habits of the capeverdean scad mackerel are not known in detail, nor are their interactions with predators higher up the food chain. For a better understanding of the status of the capeverdean scad mackerel in the food chain, it is therefore necessary to carried out detailed concurrent studies of the food and feeding habits of the capeverdean scad mackerel and its predators (such as tuna). It is also necessary to study the availability of the prey items in relation to concurrent seasonal abundance and distribution of the predator species.

All of the assessment procedures described in earlier chapters were used in order to estimate the size and the potential yield of the capeverdean scad mackerel stock. Assessment results indicate that in 1995 the spawning stock biomass was around 19 thousands tones and the fishing mortality equal to 0.15 per year. Maximum

sustainable yield has been estimated to be equal to 5 - 6 thousand tons at fishing mortality equal to 0.3 (see Figure 6.6.2). Since current catch (in 1996) is around 3 thousand tons, we must conclude that there is room for increasing the catch to a level of 3 to 5 - 6 thousand tons without putting the stock at risk. However, all the procedures and methods used herein contain uncertainties and one should interpret the results with care. It must also be kept in mind that only a "prudent exploitation strategy" can avoid the potential risk of a stock collapse.

For any exploited stock, it is of paramount importance that the means needed to assess its true abundance are available. It has been shown in this study that the CPUE data (Table 1) can be used as an index of abundance of the capeverdean scad mackerel. Hence, acoustical surveys should be carried out in order to obtain a better index of abundance of the capeverdean scad mackerel for tuning the cohort analysis.

8. ACKNOWLEDGEMENTS

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Appendix

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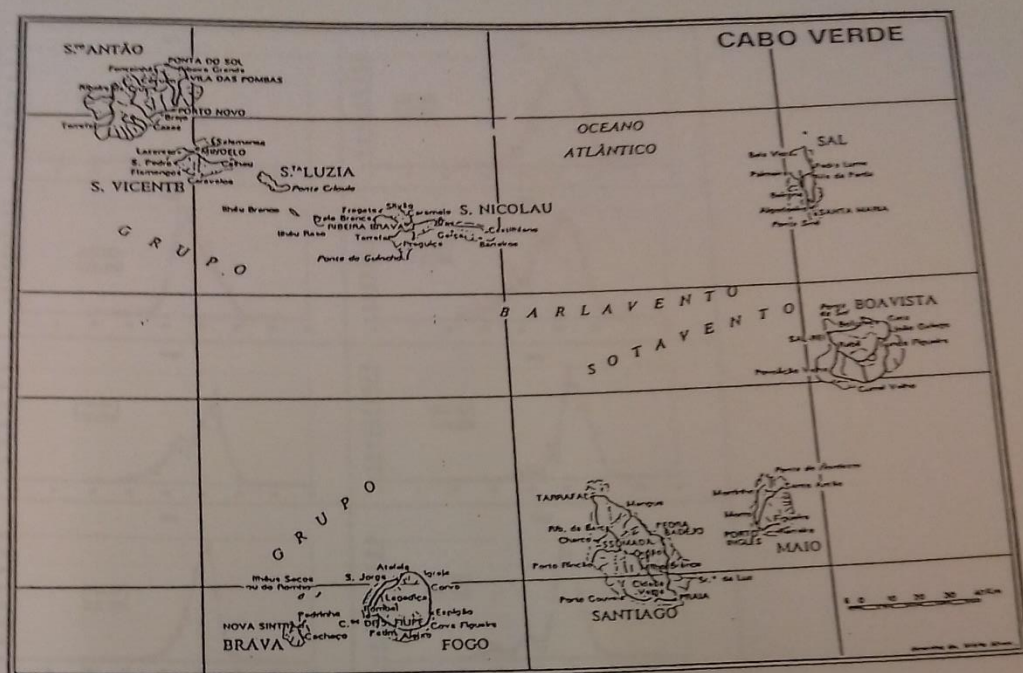


Figure 1. The archipelago of Cape Verde

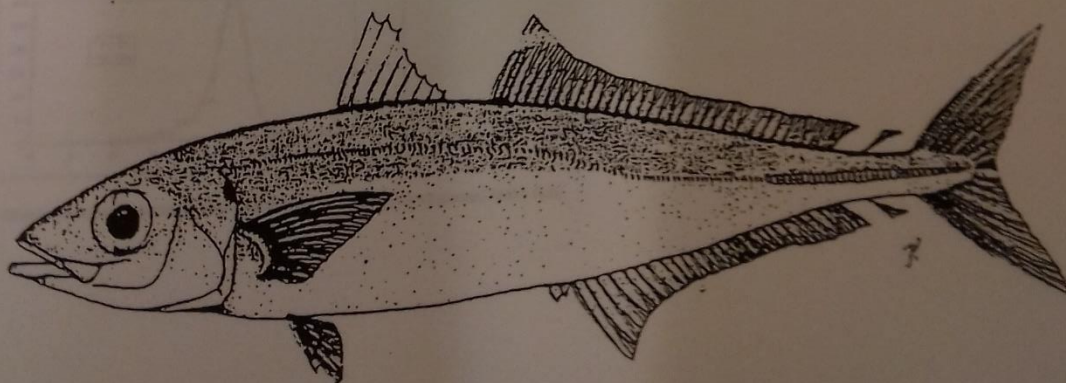


Figure 2. *Decapterus macarellus* (Cuvier, 1833)

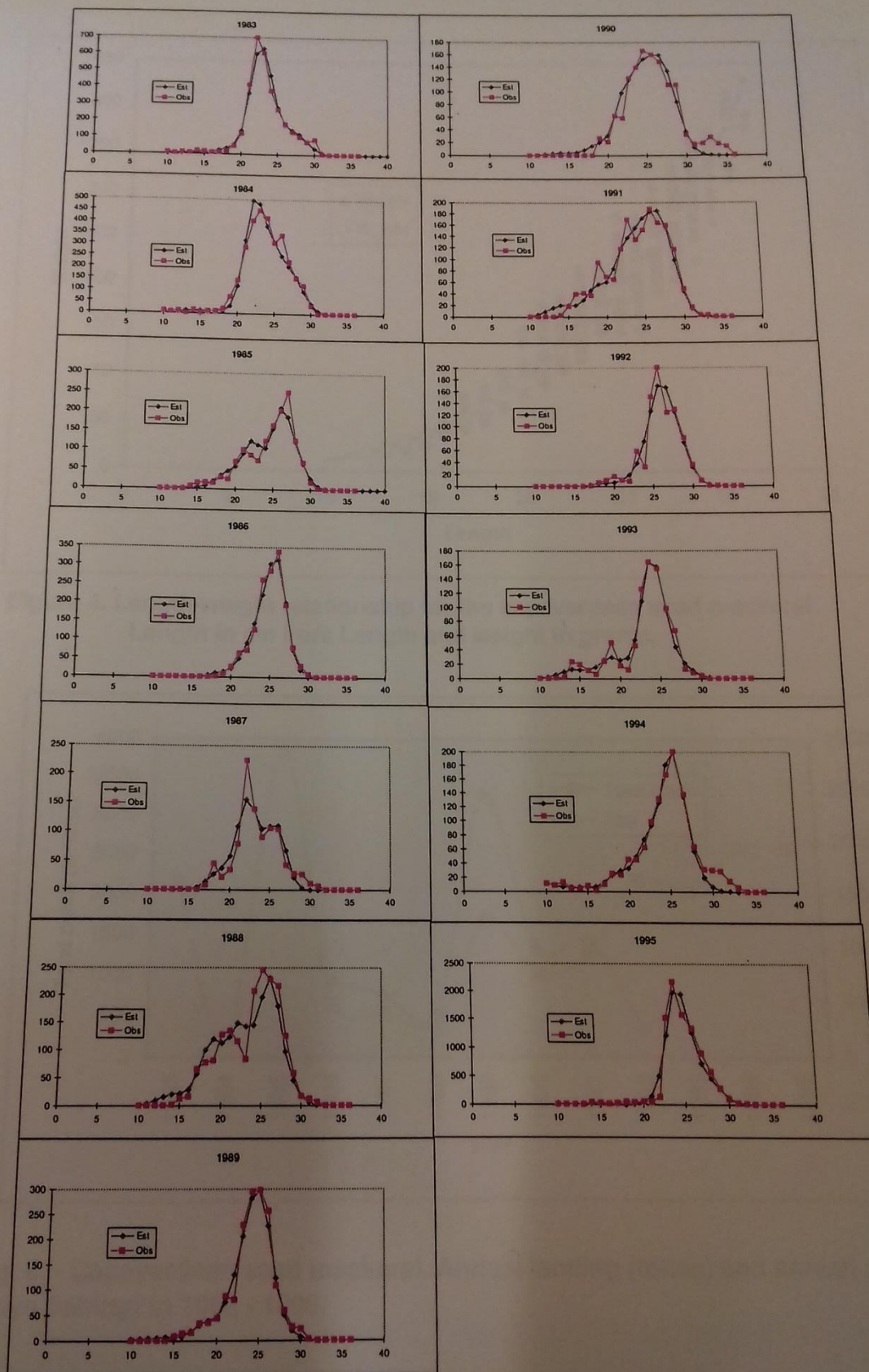


Figure 3. Capeverdean scad mackerel. Observed and estimated length-frequencies, 1983 - 1995.

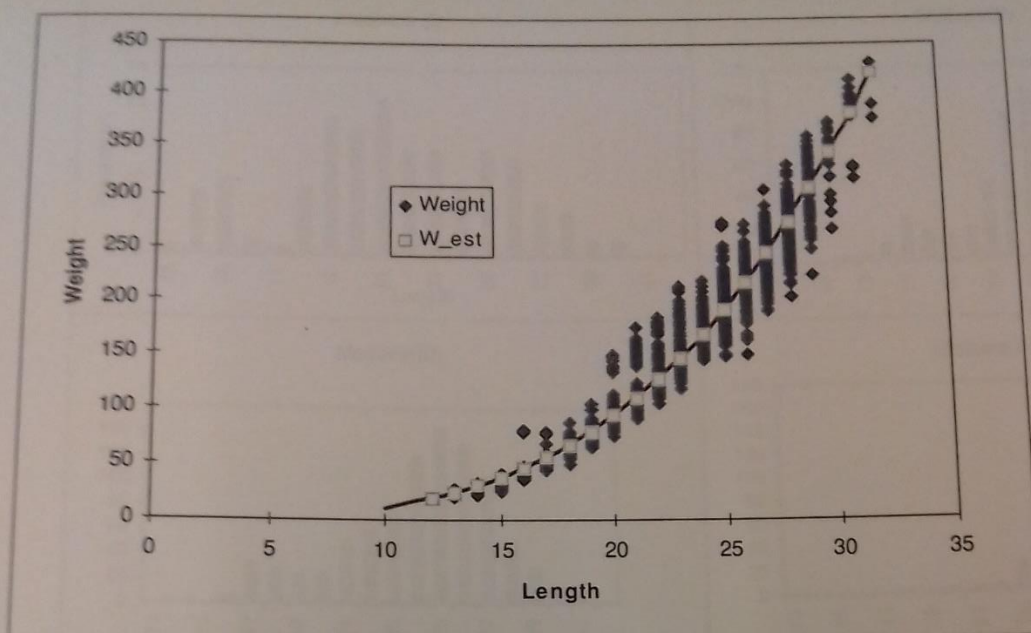


Figure 4. Length-weight relationship for the capeverdean scad mackerel.
Length in cm Fork Length and weight in grams.

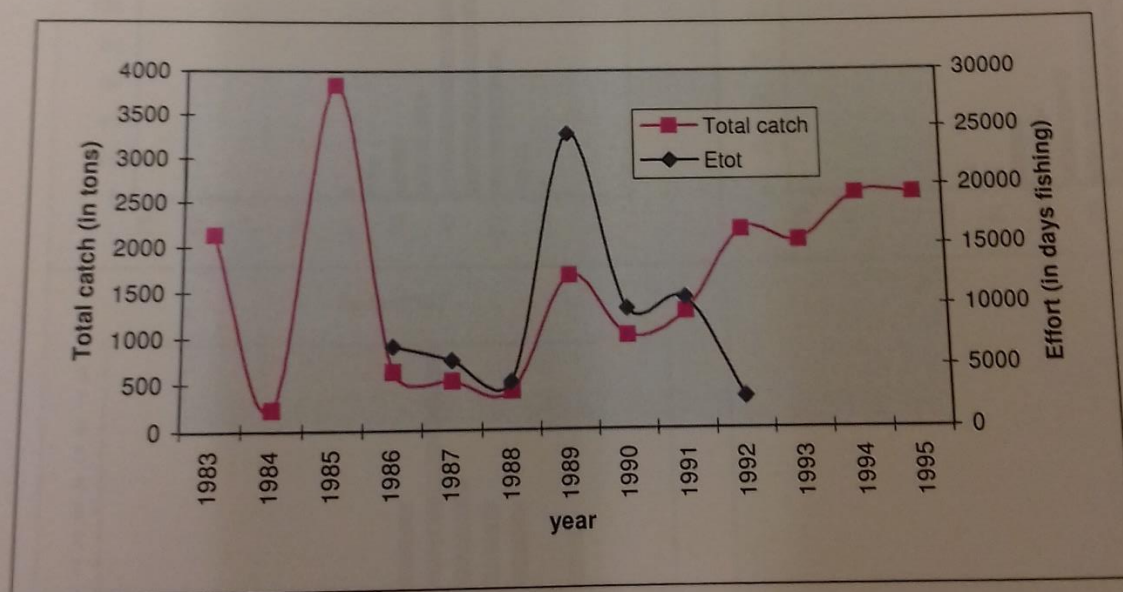


Fig. 6. - Capeverdean scad mackerel. Annual landing (tonnes) and annual standardized effort (days fishing) in 1983 - 1995.

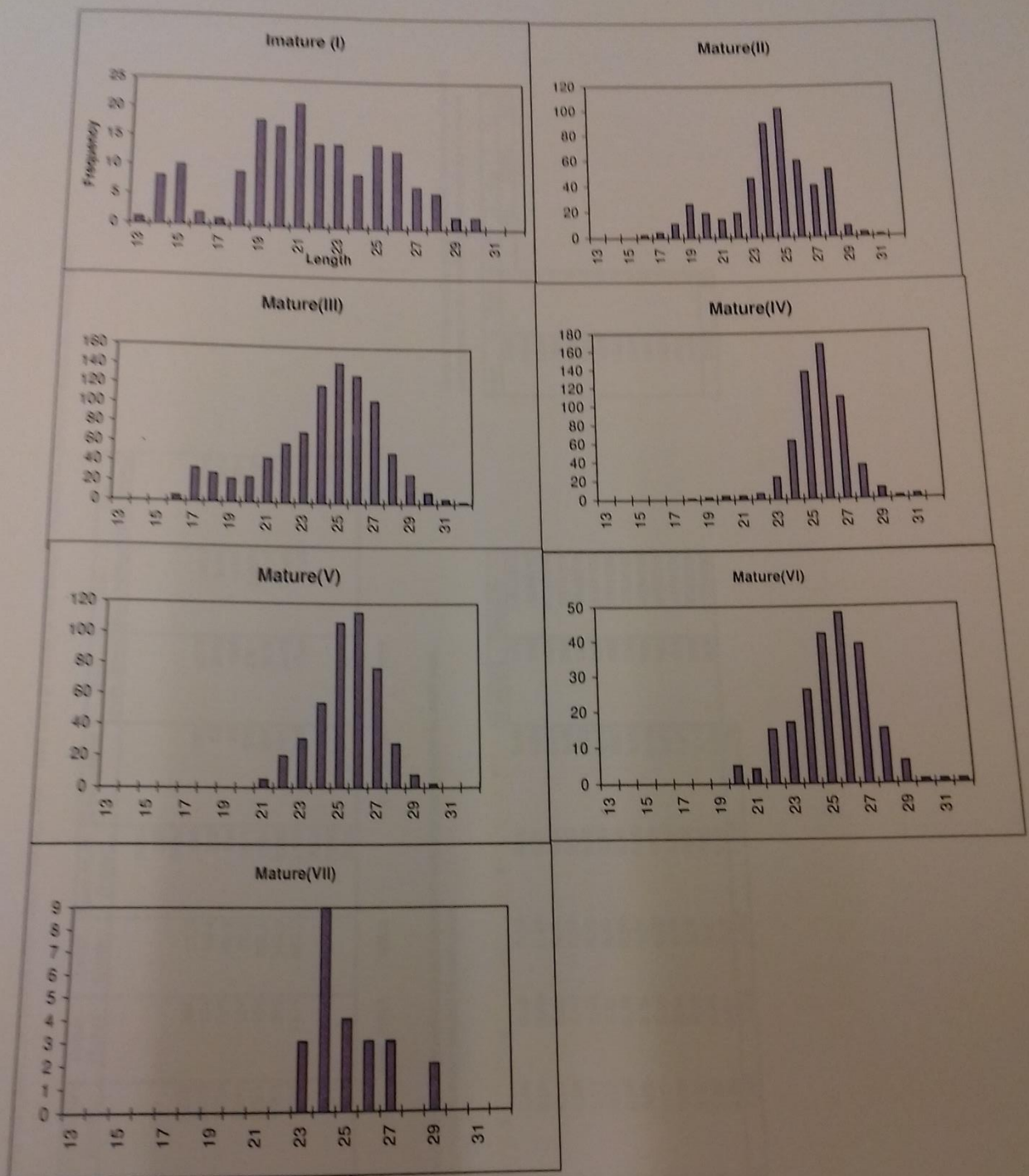


Fig. 5. Capeverdean scad mackerel. Frequency distributions of maturity stages by length

Table 1. - capeverdean scad mackerel. Annual landings (tons) and annual effort (days fishing), 1983 - 1995. Data provided by the Instituto Nacional de Desenvolvimento das Pescas (INAP)

Year	Total Effort artisan.	Catch (Tons)	CPUE kg/day	Total Effort Indust.	Catch (Tons)	CPUE kg/day	Total catch (Tons)	Artes. scaled	Eind (Days fishing)	Eart (Days fishing)	Etot (Days fishing)
1983							2134				
1984							222				
1985							3836				
1986	119851	409	3.4	2272	208	91.5		123	2272	4458	6740
1987	134478	233	1.7	2218	154	69.4	518	62	2218	3356	5574
1988	116993	229	2.0	1668	175	104.9	404	70	1668	2183	3851
1989	154358	1614	10.5	2301	167	72.5	1671	376	2301	22238	24539
1990	138070	794	5.8	2021	208	102.9	1001	207	2021	7715	9736
1991	131113	1025	7.8	2053	245	119.3	1263	281	2053	8589	10642
1992	112737	1018	9.0	1304	1151	882.7	2169	325	1304	1153	2457
1993							2545				
1994							2559				
1995							2559				

Aver_CPUE 5,7 Aver_CPUE 206,2 Aver_Eind= 1977

Table 2. Computed proportions of numbers of capeverdean scad mackerel split into age-groups for each year, catch mean weight for each year, and total number landed in each year.

Year Age	1	2	3	4	5	6	7	M_weight	Catch_N
1983	0.00	0.01	0.04	0.45	0.33	0.00	0.18	171	12314366
1984	0.02	0.02	0.01	0.46	0.25	0.01	0.24	184	1204740
1985	0.00	0.01	0.11	0.25	0.08	0.22	0.33	204	18816315
1986	0.00	0.00	0.03	0.12	0.35	0.43	0.06	208	2973278
1987	0.00	0.00	0.11	0.46	0.15	0.26	0.00	174	2360810
1988	0.00	0.05	0.22	0.21	0.12	0.23	0.16	187	2157938
1989	0.02	0.02	0.08	0.16	0.50	0.16	0.07	189	8820386
1990	0.00	0.01	0.05	0.19	0.24	0.00	0.50	240	4165766
1991	0.00	0.06	0.11	0.18	0.20	0.00	0.45	200	6330105
1992	0.00	0.00	0.02	0.04	0.15	0.17	0.62	242	8968341
1993	0.00	0.07	0.12	0.08	0.62	0.03	0.09	174	11742539
1994	0.09	0.02	0.08	0.15	0.22	0.35	0.10	205	12507629
1995	0.01	0.01	0.00	0.04	0.72	0.00	0.22	207	12366098
Std. dev	2.62	1.80	1.42	1.23	1.30	1.14	1.70		

Table 3. Capeverdean scad mackerel. Estimated of growth parameters by the method of least squares estimates.

Year	SSE	K	L _∞	t ₀
1983	3	0.34	30.06	0.11
1984	3			
1985	4			
1986	2			
1987	9			
1988	5			
1989	2			
1990	3			
1991	2			
1992	8			
1993	3			
1994	2			
1995	6			
SSE	52			

Table 4. Capeverdean scad mackerel. Estimated sexual maturity at length, based on commercial catches in the period 1988 - 1996.

Length	Prop M_t96	Prop M_mal.96	Prop M_Fem.96	Prop M_F+M [1988-1995]
10	0,00	0,00	0,00	0,01
11	0,00	0,00	0,00	0,01
12	0,00	0,00	0,00	0,02
13	0,00	0,00	0,00	0,04
14	0,00	0,00	0,00	0,06
15	0,00	0,00	0,00	0,09
16	0,00	0,00	0,00	0,14
17	0,01	0,01	0,01	0,20
18	0,02	0,03	0,03	0,29
19	0,06	0,07	0,07	0,40
20	0,14	0,16	0,18	0,51
21	0,31	0,32	0,39	0,63
22	0,55	0,55	0,64	0,73
23	0,77	0,76	0,83	0,81
24	0,90	0,89	0,93	0,87
25	0,96	0,95	0,98	0,92
26	0,99	0,98	0,99	0,95
27	1,00	0,99	1,00	0,97
28	1,00	1,00	1,00	0,98
29	1,00	1,00	1,00	0,99
30	1,00	1,00	1,00	0,99
31	1,00	1,00	1,00	0,99
32	1,00	1,00	1,00	1,00

Table 8. Capeverdean scad mackerel. Catch at age-groups in number (in millions), 1983 - 1995

Year/Age	1	2	3	4	5	6	7
1983	0.00	0.15	0.49	5.59	4.07	0.00	2.21
1984	0.02	0.02	0.02	0.56	0.30	0.01	0.28
1985	0.00	0.24	1.99	4.73	1.42	4.16	6.27
1986	0.01	0.00	0.08	0.37	1.04	1.28	0.19
1987	0.00	0.01	0.34	1.38	0.46	0.79	0.00
1988	0.00	0.11	0.48	0.46	0.26	0.49	0.36
1989	0.15	0.15	0.70	1.38	4.37	1.43	0.64
1990	0.00	0.06	0.20	0.80	1.02	0.00	2.09
1991	0.00	0.35	0.72	1.12	1.29	0.00	2.86
1992	0.00	0.04	0.19	0.32	1.39	1.49	5.54
1993	0.00	0.79	1.41	0.92	7.24	4.34	1.20
1994	1.08	0.24	1.03	1.84	2.76	4.34	1.20
1995	0.07	0.13	0.00	0.55	8.88	0.00	2.75

Table 10. Capeverdean scad mackerel. Fishing mortality at age-groups, 1983 - 1995.
The averages are taken across ages-groups 2 - 6.

Year/Age	1	2	3	4	5	6	7	AveF
1983	0.000	0.004	0.058	0.332	0.040	0.000	0.124	0.108
1984	0.000	0.001	0.001	0.106	0.032	0.000	0.150	0.035
1985	0.000	0.002	0.074	0.340	0.555	1.118	0.150	0.522
1986	0.000	0.000	0.001	0.022	0.142	6.585	0.150	1.688
1987	0.000	0.000	0.003	0.026	0.041	0.190	0.150	0.065
1988	0.000	0.002	0.002	0.006	0.007	0.070	0.150	0.021
1989	0.001	0.002	0.015	0.010	0.092	0.063	0.150	0.045
1990	0.000	0.000	0.004	0.026	0.012	0.000	0.150	0.010
1991	0.000	0.030	0.006	0.035	0.064	0.000	0.150	0.026
1992	0.000	0.025	0.004	0.004	0.069	0.122	0.150	0.055
1993	0.000	0.074	0.010	0.201	0.135	0.023	0.150	0.092
1994	0.209	2.726	0.162	0.019	5.356	0.139	0.150	1.419
1995	0.005	0.042	0.150	0.150	0.150	0.150	0.150	0.150

Table 9. Capeverdean scad mackerel. Stock size in number at age-groups (in millions), 1983 - 1995

Year/Age	1	2	3	4	5	6	7
1983	75.61	44.84	10.64	24.18	125.90	3.66	22.87
1984	220.26	50.68	29.94	6.73	11.63	81.06	2.45
1985	302.39	147.62	33.96	20.06	4.06	7.55	54.33
1986	542.34	202.69	98.76	21.13	9.57	1.56	1.65
1987	130.18	363.53	135.87	66.13	13.86	5.56	0.00
1988	131.14	87.26	243.68	90.80	43.20	8.92	3.08
1989	349.60	87.90	58.41	162.95	60.49	28.74	5.57
1990	21.70	234.22	58.80	38.58	108.10	36.97	18.09
1991	392.59	14.55	156.95	39.25	25.21	71.62	24.78
1992	20.19	263.16	9.46	104.62	25.40	15.84	48.01
1993	0.48	13.53	176.37	6.19	69.87	15.89	9.40
1994	7.01	0.32	8.42	117.07	3.39	40.91	10.41
1995	18.36	3.81	0.01	4.80	76.97	0.01	23.86

Table 11. Capeverdean scad mackerel. The biomass at age-groups in '000 t, the total and the SSB in '000 t, 1983 - 1995.

Year/Age	1	2	3	4	5	6	7	Total	SSB
1983	0.46	1.90	0.80	3.14	20.16	0.80	5.79	33.06	30.39
1984	1.35	2.15	2.26	0.87	1.86	17.70	0.62	26.81	22.11
1985	1.85	6.25	2.57	2.60	0.65	1.65	13.75	29.32	20.28
1986	3.32	8.58	7.47	2.74	1.53	0.34	0.42	24.40	8.67
1987	0.80	15.39	10.27	8.59	2.22	1.21	0.00	38.48	17.41
1988	0.80	3.69	18.42	11.79	6.92	1.95	0.78	44.35	28.39
1989	2.14	3.72	4.42	21.15	9.69	6.28	1.41	48.80	40.52
1990	0.13	9.92	4.45	5.01	17.31	8.07	4.58	49.46	37.67
1991	2.40	0.62	11.86	5.10	4.04	15.64	6.27	45.93	35.34
1992	0.12	11.14	0.72	13.58	4.07	3.46	12.15	45.24	34.77
1993	0.00	0.57	13.33	0.80	11.19	3.47	2.38	31.75	22.64
1994	0.04	0.01	0.64	15.20	0.54	8.93	2.63	28.00	27.54
1995	0.11	0.16	0.00	0.62	12.33	0.00	6.04	19.27	19.01

Table 12. Capeverdean scad mackerel. Estimated yield at age-groups in '000 t, 1983 - 1995.

Year/age	1	2	3	4	5	6	7	total
1983	0,00	0,01	0,04	0,73	0,65	0,00	0,56	1,98
1984	0,00	0,00	0,00	0,07	0,05	0,00	0,07	0,20
1985	0,00	0,01	0,15	0,61	0,23	0,91	1,59	3,50
1986	0,00	0,00	0,01	0,05	0,17	0,28	0,05	0,55
1987	0,00	0,00	0,03	0,18	0,07	0,17	0,00	0,45
1988	0,00	0,00	0,04	0,06	0,04	0,11	0,09	0,34
1989	0,00	0,01	0,05	0,18	0,70	0,31	0,16	1,41
1990	0,00	0,00	0,02	0,10	0,16	0,00	0,53	0,81
1991	0,00	0,01	0,05	0,14	0,21	0,00	0,72	1,14
1992	0,00	0,00	0,01	0,04	0,22	0,33	1,40	2,01
1993	0,00	0,03	0,11	0,12	1,16	0,06	0,27	1,76
1994	0,01	0,01	0,08	0,24	0,44	0,95	0,30	2,03
1995	0,00	0,01	0,00	0,07	1,42	0,00	0,70	2,20

Table 13. Estimated mean length at age (L), mean weight at age (W), maturity ratio (P) at age, and selection pattern at age (S).

Age	L bar	W bar	P	S
1	8,44	0,01	0,00	0,28
2	15,52	0,04	0,11	1,00
3	18,63	0,08	0,36	1,00
4	22,09	0,13	0,74	1,00
5	23,60	0,16	0,85	1,00
6	26,02	0,22	0,95	1,00
7	27,26	0,25	0,97	1,00